

5G RAN

Flexible User Steering Feature Parameter Description

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for RFSP-based flexible configuration of inter-RAT handover measurement parameters for voice and data service UEs. For details, see 4.1.2.2.1 Flexible Configuration of Handover Measurement Parameters for Voice and Data Service UEs .	Added parameters: gNBRfspConf <i>ig.DataInterRatHoGroupId</i> gNBRfspConf <i>ig.VolInterRatHoGroupId</i> gNBRfspConf <i>ig.DataInterRatHoEutraMeasGrpId</i> gNBRfspConf <i>ig.VolInterRatHoEutraMeasGrpId</i>	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 02 (2025-04-27).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for flexible configuration of threshold for event A4 and threshold 2 for event A5 in frequency-priority-based inter-frequency handover scenarios. For details, see 4.1.2.2.4 Flexible Configuration in Frequency-Priority-based Inter-Frequency Handover Scenarios .	Added the gNB RfSP Config. <i>FreqPriIFHoA4RsrpThldOfs</i> parameter.	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added support for flexible configuration of Freq Priority HO Trigger Mode Based on RFSP . For details, see 4.1.2.2.4 Flexible Configuration in Frequency-Priority-based Inter-Frequency Handover Scenarios .	Added the gNB RfSP Config. <i>RfSPFreqPriHoTriggerMode</i> parameter.	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added support for flexible configuration of threshold for event A4 and threshold 2 for event A5 in service-based inter-frequency handover scenarios. For details, see 4.1.2.2.5 Flexible Configuration in Service-based Inter-Frequency Handover Scenarios .	Added the gNB RfSP Config. <i>ServBasedInterFHoA4RsrpOfs</i> parameter.	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for RFSP-based flexible configuration of handover measurement parameters for voice and data service UEs. For details, see 4.1.2.2.1 Flexible Configuration of Handover Measurement Parameters for Voice and Data Service UEs .	Added parameters: • gNBRfspConfig.VoInterFreqHoGroupId • gNBRfspConfig.DataInterFreqHoGroupId • gNBRfspConfig.VoIntraFreqHoGroupId • gNBRfspConfig.DataIntraFreqHoGroupId	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added support for flexible configuration of thresholds for events A4 and A5 of voice service UEs in the policy of RFSP-based inter-frequency handover back to the HPLMN. For details, see 4.1.2.3 Policy of RFSP-based Inter-Frequency Handover Back to the HPLMN .	Added parameters: • gNBRfspConfig.VoBackHplmnA5RsrpThreshold1Ofs • gNBRfspConfig.VoBackHplmnA5RsrpThreshold2Ofs	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

Moved descriptions of the principles for the gNodeB to deliver frequencies to VoNR-incapable UEs of the home operator in roaming scenarios from [4.1.1 RFSP-based Cell Reselection Policy](#) to [4.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FBFD-050103	Flexible User Steering	4 RFSP-based Camping and Handover Policies 5 RFSP-based DRX Policy 6 RFSP-based VoNR Policy

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
RFSP-based camping and handover policies	The RFSP-based cell reselection policy applies only to SA networking. The RFSP-based handover policy applies to NSA and SA networking. In this case, handover measurement parameters can be flexibly configured for voice and data service UEs based on RFSPs only in SA networking.	4 RFSP-based Camping and Handover Policies
RFSP-based DRX policy	None	5 RFSP-based DRX Policy
RFSP-based VoNR policy	The RFSP-based VoNR policy applies only to SA networking in that VoNR does not apply to NSA networking.	6 RFSP-based VoNR Policy

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
RFSP-based camping and handover policies	High frequency bands support only SA networking in FWA scenarios, while low frequency bands support SA networking in all scenarios. The RFSP-based unnecessary handover policy applies only to low-frequency scenarios.	4 RFSP-based Camping and Handover Policies
RFSP-based DRX policy	In low-frequency scenarios, this function applies to both SA and NSA networking. In high-frequency scenarios, this function applies only to NSA networking.	5 RFSP-based DRX Policy
RFSP-based VoNR policy	None	6 RFSP-based VoNR Policy

3 Overview

If the same radio resource management (RRM) policy is used for all UEs served by a gNodeB on a radio network, the requirements of UEs performing different services and those with different load status cannot be met.

To provide specific RRM policies for UEs, subscriber profile IDs (SPIDs) are introduced in LTE, and RAT/Frequency Selection Priorities (RFSPs) are introduced in NR. An operator registers SPIDs (ranging from 1 to 256) for UEs with the home subscriber server (HSS) database and registers RFSPs (ranging from 1 to 256) for UEs with the unified data management (UDM) database.

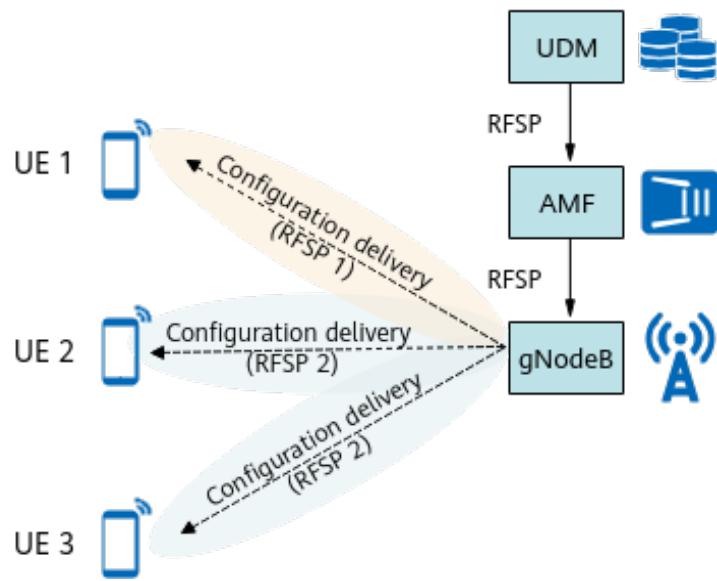
Flexible User Steering

The Flexible User Steering feature groups UEs with the same characteristics by RFSP or SPID. This enables the gNodeB to flexibly customize RRM policies (such as camping, handover, and DRX) for UEs in different groups, improving service experience.

- In SA networking, the access and mobility management function (AMF) sends the RFSP indexes to a gNodeB over the NG interface. After mapping the RFSP indexes to the local configuration, the gNodeB provides specific RRM policies for UEs. In this way, targeted services can be better provided based on service characteristics and load status of UEs.
- In NSA networking, the evolved packet core (EPC) sends the SPID indexes to the eNodeB over the S1 interface, and the eNodeB sends the SPID indexes to the gNodeB through the *subscriberProfileIDforRFP* IE in the SgNB Addition Request message. After mapping the SPID indexes to the local configuration, the gNodeB provides specific RRM policies for UEs based on the indexes. In this way, targeted services can be better provided based on service characteristics and load status of UEs. It can be noted that in NSA networking, Flexible User Steering is implemented using SPIDs in LTE, which are equivalent to RFSPs in NR.

In NSA networking, after the eNodeB sends SPIDs to the gNodeB, the gNodeB processes the SPIDs in the same way as it processes the RFSPs in SA networking. This document uses SA networking as an example to describe the Flexible User Steering feature. [Figure 3-1](#) shows how to use RFSPs to flexibly customize UE policies in SA networking.

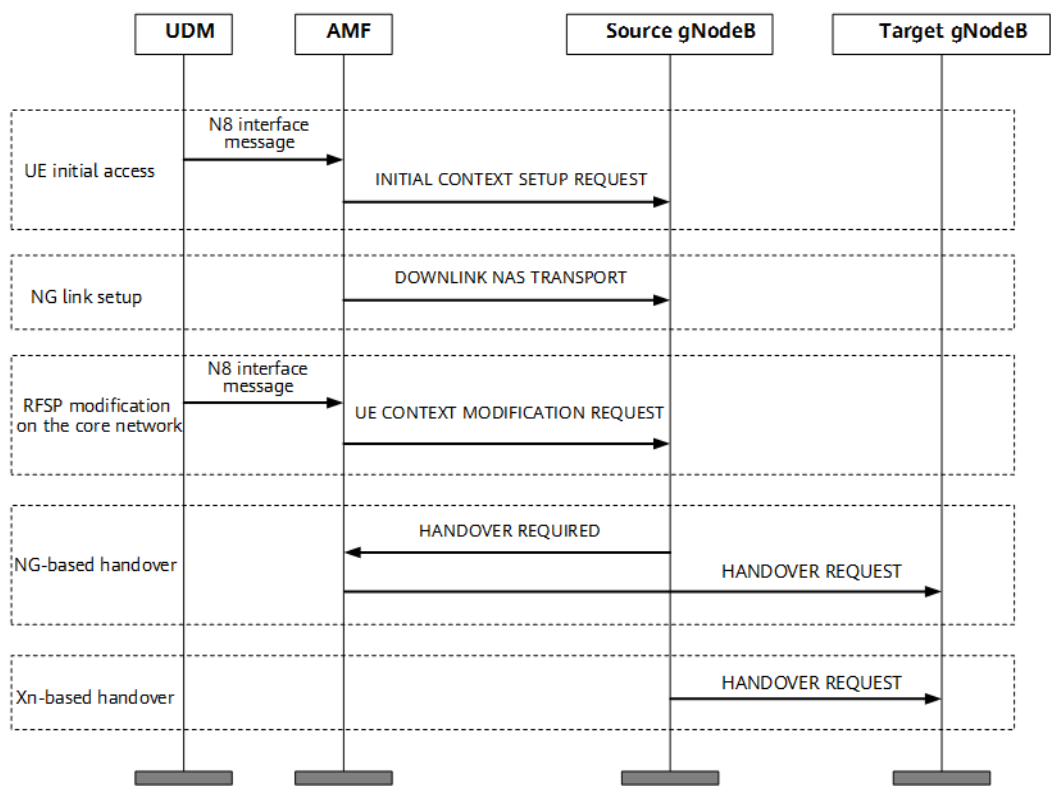
Figure 3-1 Flexible user steering



Transmission of an RFSP on the Network

An RFSP is transmitted over different signaling messages in different phases, as shown in Figure 3-2.

Figure 3-2 Transmission of an RFSP on the network



- UE initial access

The UDM sends an RFSP configured by the operator to the AMF over the N8 interface message. The AMF forwards the RFSP to the gNodeB using an INITIAL CONTEXT SETUP REQUEST message.

- NG link setup
If the UE migrates from idle mode to connected mode, the AMF sends a DOWNLINK NAS TRANSPORT message containing the RFSP received from the UDM to the gNodeB.
- RFSP modification on the core network
During data transmission, if the UDM modifies the RFSP, the UDM sends the updated RFSP to the AMF over the N8 interface message. The AMF then sends the updated RFSP to the gNodeB using a UE CONTEXT MODIFICATION REQUEST message.
- NG-based handover
The source gNodeB sends a HANDOVER REQUIRED message containing an RFSP to the AMF. The AMF then sends a HANDOVER REQUEST message containing the RFSP to the target gNodeB.
- Xn-based handover
The source gNodeB sends a HANDOVER REQUEST message containing the RFSP obtained from the AMF to the target gNodeB.

4 RFSP-based Camping and Handover Policies

4.1 Principles

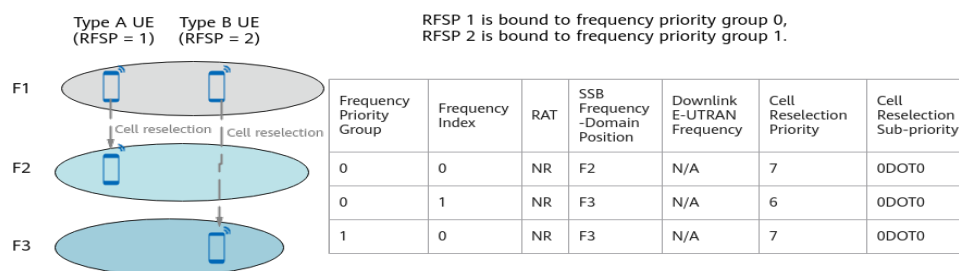
In inter-frequency or inter-RAT networking scenarios, operators expect user-level mobility control, which allows specific UEs to preferentially reselect or be handed over to a specific network. In this context, RFSP-based camping and handover policies are introduced. [Table 4-1](#) describes the sub-functions and corresponding typical application scenarios.

Table 4-1 Sub-functions and application scenarios

Sub-Function	Application Scenario
4.1.1 RFSP-based Cell Reselection Policy	Operators expect specific UEs to camp on specific frequencies or cells. For example, customer-premises equipment (CPE) users are expected to preferentially camp on cells operating on specified frequencies.
4.1.2 RFSP-based Handover Policy	Operators expect specific UEs to be handed over to specific frequencies or cells. For example, UEs that do not support Voice over New Radio (VoNR) must not be handed over to areas with only NR coverage but no LTE coverage, preventing the situation where UEs can receive signals but fail to set up calls.

4.1.1 RFSP-based Cell Reselection Policy

The RFSP-based cell reselection policy controls whether specific UEs camp on the cell operating on a specific frequency of a certain RAT (NG-RAN or E-UTRAN) preferentially based on RFSP-specific dedicated priorities during cell reselection, as shown in [Figure 4-1](#).

Figure 4-1 RFSP-based cell reselection policy**NOTE**

This document describes only the content exclusive to RFSP-based cell reselection compared with basic cell reselection functions. For details about basic cell reselection functions, see *SA Mobility Management in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

The following describes how RFSP-specific dedicated priorities for cell reselection are configured and delivered on the gNodeB side as well as how UEs use these priorities for cell reselection:

1. Configure frequency priority groups on the gNodeB side.

The **gNBFreqPriorityGroup** MO is used to configure gNodeB frequency priority groups. [Table 4-2](#) lists the key parameters.

Table 4-2 gNodeB frequency priority group

Parameter Name	Parameter ID	Description
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupID	This parameter specifies the ID of a gNodeB frequency priority group.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	This parameter specifies the index of a frequency in the gNodeB frequency priority group.
RAT Type	gNBFreqPriorityGroup.RatType	This parameter specifies the RAT corresponding to the configured frequency.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	This parameter specifies the SSB frequency-domain position of an NR frequency.
Downlink EARFCN	gNBFreqPriorityGroup.DLEarfcn	This parameter specifies the downlink frequency of a neighboring E-UTRAN cell.

Parameter Name	Parameter ID	Description
Cell Reselection Priority	gNBFreqPriorityGroup.CellReselPri	This parameter specifies the cell reselection priority of the configured frequency. A larger value of this parameter indicates a higher priority.
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPri	This parameter specifies the cell reselection sub-priority of the configured frequency. A larger value of this parameter indicates a higher priority. If this parameter is set to 0DOT0 , the cell reselection sub-priority is not delivered.
Note: The UE adds up the cell reselection priority and the cell reselection sub-priority of a configured frequency and uses the sum as the final cell reselection priority of the frequency.		

Table 4-3 lists configuration examples of gNodeB frequency priority groups.

Table 4-3 Configuration examples of gNodeB frequency priority groups

gNodeB Frequency Priority Group ID	Frequency Index	RAT Type	SSB Frequency Position	Downlink EARFCN	Cell Reselection Priority	Cell Reselection Sub-Priority
0	0	NR	F1	N/A	5	0DOT2
0	1	E-UTRAN	N/A	F3	4	0DOT2
1	0	NR	F1	N/A	5	0DOT2
1	1	NR	F2	N/A	4	0DOT2
1	2	E-UTRAN	N/A	F4	3	0DOT2
1	3	E-UTRAN	N/A	F5	3	0DOT2
Note: F1, F2, F3, F4, and F5 in the table exemplify different frequencies.						

2. The gNodeB obtains the RFSP indexes of UEs from the core network.
3. Based on the obtained RFSP indexes, bind the RFSP indexes to the configured gNodeB frequency priority groups on the gNodeB to generate the RFSP-specific dedicated priorities.

The **gNBRfspConfig** MO is used for the binding. **Table 4-4** lists the related parameters.

Table 4-4 Parameters for binding an RFSP index to a gNodeB frequency priority group

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.
gNodeB Frequency Priority Group ID	gNBRfspConfig.gNBFreqPriorityGroupId	This parameter specifies the ID of the gNodeB frequency priority group bound to the RFSP index. This parameter has a mapping relationship with the gNBFreqPriorityGroup.gNBFreqPriorityGroupId parameter. If this parameter is set to 255, the gNodeB frequency priority group bound to the RFSP index is invalid.

NOTE

In RedCap/eRedCap scenarios, the gNodeB frequency priority group ID bound to the RFSP index is specified by the **gNBRfspConfig.RedCapGnbFreqPriGroupId** parameter. For details, see the section "Frequency Selection for RedCap" in *RedCap*.

4. The gNodeB determines whether RFSP-based cell reselection takes effect for the UE corresponding to the RFSP index.
 - If the RFSP index is bound to a valid gNodeB frequency priority group, RFSP-based cell reselection takes effect for the UE. In this case, the process proceeds to 5.
 - If the RFSP index is bound to an invalid gNodeB frequency priority group, RFSP-based cell reselection does not take effect for the UE. In this case, the process ends.
5. The gNodeB delivers the RFSP-specific dedicated frequency priorities to the UE through an RRCRelease message.
 - a. The gNodeB uses the configured RFSP frequency list and corresponding frequency priorities to generate a frequency priority list.
 - b. The gNodeB selects the frequencies based on the UE's frequency support capability and its target public land mobile network (PLMN).
 - c. The gNodeB includes the selected frequencies in the *cellReselectionPriorities* IE of an RRCRelease message and sends the message to the UE.

In low-frequency scenarios, the gNodeB also complies with the following rules when delivering frequencies for CA-capable UEs in CA scenarios:

If the **CA_UE_FREQ_STEER_SW** option of the **NRCCellAlgoSwitch.MultiFreqAlgoSwitch** parameter is selected in the serving cell, the gNodeB determines whether to filter out the neighboring frequencies based on CA capabilities of UEs, the frequency of the serving cell on which the UEs camp, and networking continuity. The frequencies that are not filtered out are delivered to the UEs, and the UEs can reselect the corresponding frequencies based on the RFSP-specific dedicated priorities.

For details about frequency filtering scenarios and recommended scenarios for CA UE frequency steering, see *SA Mobility Management in Connected Mode*.

NOTE

- According to 3GPP TS 38.304, the frequency specified by the **gNBFreqPriorityGroup.SsbFreqPos** parameter in the intra-RAT frequency priority list must have been delivered in SIB4, and the frequency specified by the **gNBFreqPriorityGroup.DLEarfcn** parameter in the inter-RAT frequency priority list must have been delivered in SIB5. If a frequency has not been delivered in SIB4 or SIB5, a UE will not select this frequency as the target frequency for cell reselection even if the frequency is contained in the *cellReselectionPriorities* IE of the RRCRelease message sent to the UE.
6. The UE obtains the RFSP-specific dedicated priorities and checks its RFSP index in the RFSP-specific dedicated priorities to determine whether RFSP-based cell reselection takes effect.
- If the RFSP index of the UE has a corresponding **gNBRfspConfig.RfspIndex**, the UE uses the gNodeB frequency priority group bound to the RFSP index for RFSP-based cell reselection.
 - If the RFSP index of the UE does not have a corresponding **gNBRfspConfig.RfspIndex**, the UE checks the configuration of RFSP index 0 in the RFSP-specific dedicated priorities.
 - If there is a **gNBRfspConfig.RfspIndex** whose value is set to **0** and a valid gNodeB frequency priority group is bound to it, the UE directly uses the gNodeB frequency priority group bound to RFSP index 0 for RFSP-based cell reselection.
 - Otherwise, RFSP-based cell reselection does not take effect on the UE.

4.1.2 RFSP-based Handover Policy

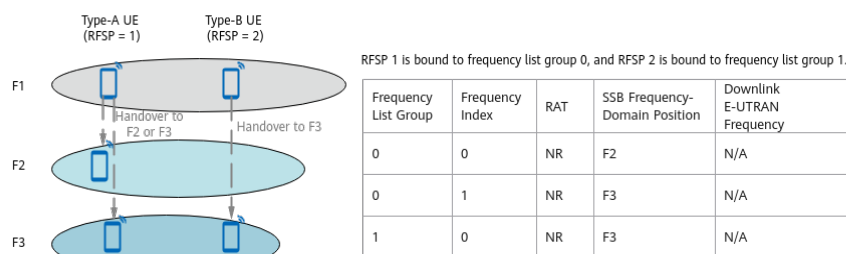
4.1.2.1 Basic RFSP-based Handover Policy

As shown in [Figure 4-2](#), the basic RFSP-based handover policy controls whether the RFSP-specific dedicated frequency list is used for handing over specific UEs to cells operating on specific frequencies of a certain RAT (NG-RAN or E-UTRAN). Cell handovers based on the RFSP-specific dedicated frequency list include:

- When an inter-frequency or inter-RAT handover is required for a UE, the gNodeB delivers only frequencies in the RFSP-specific dedicated frequency list to the UE.

- When a blind handover is required for a UE, the gNodeB hands the UE over only to a cell operating on a frequency in the RFSP-specific dedicated frequency list.
- When a redirection is required for a UE, the gNodeB redirects the UE only to a cell operating on a frequency in the RFSP-specific dedicated frequency list.

Figure 4-2 Basic RFSP-based handover policy



NOTE

This document describes the content exclusive to RFSP-based handover compared with basic cell handover functions. For details about basic cell handover functions, see *SA Mobility Management in Connected Mode* and *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

The basic RFSP-specific handover policy consists of the following sub-functions, which do not take effect for UEs performing emergency calls:

- RFSP-based NR inter-frequency handover policy: controlled by the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter
- RFSP-based NG-RAN-to-E-UTRAN handover policy: controlled by the **NR_TO_EUTRAN_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

The following describes how the RFSP-specific dedicated frequency lists for cell handovers are configured and delivered to UEs on the gNodeB side as well as how an RFSP-specific dedicated frequency list is used for handovers:

1. Configure frequency list groups on the gNodeB side.

The **gNBFreqPriorityGroup** MO is used to configure a gNodeB frequency list group for cell handovers. [Table 4-5](#) lists the key parameters.

Table 4-5 gNodeB frequency priority group

Parameter Name	Parameter ID	Description
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	This parameter specifies the ID of a gNodeB frequency priority group.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	This parameter specifies the index of a frequency in the gNodeB frequency priority group.

Parameter Name	Parameter ID	Description
RAT Type	gNBFreqPriorityGroup.RatType	This parameter specifies the RAT corresponding to the configured frequency.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	This parameter specifies the SSB frequency-domain position of an NR frequency.
Downlink EARFCN	gNBFreqPriorityGroup.DLEarfcn	This parameter specifies the downlink frequency of a neighboring E-UTRAN cell.
Traffic Priority	gNBFreqPriorityGroup.TrafficPriority	This parameter specifies the traffic priority of a non-serving NR frequency.

Table 4-6 lists configuration examples of gNodeB frequency list groups.

Table 4-6 Configuration examples of gNodeB frequency list groups

gNodeB Frequency Priority Group ID	Frequency Index	RAT Type	SSB Frequency Position	Traffic Priority	Downlink EARFCN
0	0	NR	F1	0	N/A
0	1	E-UTRAN	N/A	N/A	F3
1	0	NR	F1	1	N/A
1	1	NR	F2	2	N/A
1	2	E-UTRAN	N/A	N/A	F4
1	3	E-UTRAN	N/A	N/A	F5

Note: F1, F2, F3, F4, and F5 in the table exemplify different frequencies.

- The gNodeB obtains the RFSP indexes of UEs from the core network.
- Based on the obtained RFSP indexes, bind the RFSP indexes to the configured gNodeB frequency list groups on the gNodeB to generate the RFSP-specific dedicated frequency lists.

The **gNBRfspConfig** MO is used for the binding. **Table 4-7** lists the related parameters.

Table 4-7 Parameters for binding an RFSP index to a gNodeB frequency list group

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.
gNodeB Frequency Priority Group ID	gNBRfspConfig.gNBFreqPriorityGroupId	This parameter specifies the ID of the gNodeB frequency list group bound to the RFSP index. This parameter has a mapping relationship with the gNBFreqPriorityGroup.gNBFreqPriorityGroupId parameter. If this parameter is set to 255 , the gNodeB frequency list group bound to the RFSP index is invalid.

 NOTE

In RedCap/eRedCap scenarios, the gNodeB frequency list group ID bound to the RFSP index is specified by the **gNBRfspConfig.RedCapGnbFreqPriGroupId** parameter. For details, see the section "Frequency Selection for RedCap" in *RedCap*.

4. The gNodeB determines whether RFSP-based cell handover takes effect for the UE corresponding to the RFSP index.
 - If the RFSP index is bound to a valid gNodeB frequency list group, RFSP-based cell handover takes effect for the UE. In this case, the process proceeds to 5.
 - If the RFSP index is bound to an invalid gNodeB frequency list group, RFSP-based cell handover does not take effect for the UE. In this case, the process ends.
5. The gNodeB delivers the RFSP-specific dedicated frequency list to a UE through the RRCReconfiguration message.
6. The UE obtains the RFSP-specific dedicated frequency list and checks its RFSP index in the RFSP-specific dedicated frequency list to determine whether RFSP-based cell handover takes effect.
 - If the RFSP index of the UE has a corresponding **gNBRfspConfig.RfspIndex**, the UE uses the gNodeB frequency priority group bound to the RFSP index for RFSP-based cell handover.
 - If the RFSP index of the UE does not have a corresponding **gNBRfspConfig.RfspIndex**, the gNodeB checks the configuration of RFSP index 0 in the RFSP-specific dedicated frequency list.

- If there is a **gNBRfspConfig.RfspIndex** whose value is set to **0** and a valid gNodeB frequency list group is bound to it, the UE directly uses the gNodeB frequency list group bound to RFSP index 0 for RFSP-based cell handover.
- Otherwise, RFSP-based cell handover does not take effect on the UE.

4.1.2.2 RFSP-based Flexible Configuration of Measurement Parameters

4.1.2.2.1 Flexible Configuration of Handover Measurement Parameters for Voice and Data Service UEs

Voice and data service UEs require different handover measurement parameters. Handover measurement parameters can be flexibly configured for voice and data service UEs based on RFSPs. In this context, voice service UEs refer to UEs performing VoNR services, while data service UEs refer to those not performing VoNR services. "Flexible configuration of handover measurement parameters for voice and data service UEs" applies only to SA networking.

This function involves the following two parts:

- Flexible configuration of intra-RAT handover measurement parameters for voice and data service UEs: The intra-RAT handover refers to a handover between NR cells. For details, see [Intra-RAT Handover Scenario](#).
- Flexible configuration of inter-RAT handover measurement parameters for voice and data service UEs: The inter-RAT handover refers to a handover from an NR cell to an LTE cell. For details, see [Inter-RAT Handover Scenario](#).

"Flexible configuration of handover measurement parameters for voice and data service UEs" is implemented as follows:

For a specific RFSP index, the RFSP-specific handover measurement parameter group ID can be configured using the parameters in the **gNBRfspConfig** MO. The handover measurement parameters associated with the parameter group ID are the handover measurement parameters associated with the general handover measurement parameter group ID that has the same value as the RFSP-specific handover measurement parameter group ID.

Take the intra-RAT inter-frequency handovers for voice service UEs as an example. The RFSP-specific handover measurement parameter group ID is configured using the **gNBRfspConfig.VolInterFreqHoGroupId** parameter. If this ID is set to **1**, the handover measurement parameters used for RFSP-specific voice service UEs during inter-frequency handovers are those whose **NRCeIlInterFHoMeaGrp.InterFreqHoMeasGroupId** (general handover measurement parameter group ID) is also **1**.

If the RFSP-specific handover measurement parameter group ID is set to **255** or there is no general handover measurement parameter group ID with the same value in the serving cells of the UEs, the RFSP-specific handover measurement parameter group ID does not take effect.

Intra-RAT Handover Scenario

Table 4-8 Parameters of intra-RAT handover measurement parameter group IDs for different types of UEs in different scenarios

Scenario	UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Inter-frequency	Voice service UE	gNBRfspConfig.VolInterFreqHoGroupId	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId
Inter-frequency	Data service UE	gNBRfspConfig.DataInterFreqHoGroupId	NRCellInterFHoMeaGrp.InterFreqHoMeasGroupId
Intra-frequency	Voice service UE	gNBRfspConfig.VolIntraFreqHoGroupId	NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId
Intra-frequency	Data service UE	gNBRfspConfig.DataIntraFreqHoGroupId	NRCellIntraFHoMeaGrp.IntraFreqHoMeasGroupId
Note: For details about the general handover measurement parameter group ID and its involved handover measurement parameters in intra-RAT handover scenarios, see <i>SA Mobility Management in Connected Mode</i> .			

Inter-RAT Handover Scenario

Table 4-9 Parameters of inter-RAT handover measurement parameter group IDs for different types of UEs

UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Voice service UE	gNBRfspConfig.VolInterRatHoGroupId	NRCellInterRHoMeaGrp.InterRatHoMeasGroupId

UE Type	RFSP-specific Handover Measurement Parameter Group ID Parameter	General Handover Measurement Parameter Group ID Parameter
Voice service UE	gNB RfSP Config.VoInterRHoEutraMeasGrpId	NRCellHoEutranMeasGrp.InterRHoToEutranMeasGrpId
Data service UE	gNB RfSP Config.DataInterRatHoGroupId	NRCellInterRHoMeasGrp.InterRatHoMeasGroupId
Data service UE	gNB RfSP Config.DataInterRHoEutraMeasGrpId	NRCellHoEutranMeasGrp.InterRHoToEutranMeasGrpId
Note 1: For details about the general handover measurement parameter group ID and its involved handover measurement parameters in inter-RAT handover scenarios, see <i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>. Note 2: For voice service UEs, the ranges of handover measurement parameters corresponding to the gNB RfSP Config.VoInterRatHoGroupId and gNB RfSP Config.VoInterRHoEutraMeasGrpId parameters are different. The differences are similar to those between the parameters corresponding to the general handover measurement parameter group ID. The same rules apply to data service UEs.		

4.1.2.2.2 Flexible Configuration in Coverage-based Inter-Frequency Handover Scenarios

In intra-RAT RFSP-specific coverage-based inter-frequency handover scenarios, if the RFSP-specific coverage-based inter-frequency handover function is enabled, thresholds for events A1, A2, and A5 can be customized for UEs based on RFSPs. The RFSP-specific coverage-based inter-frequency handover function is controlled by the **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

The gNodeB selects the thresholds for events A1, A2, and A5 based on whether the RFSP-specific coverage-based inter-frequency handover function is enabled. In addition, threshold 1 for event A5 and thresholds for events A1 and A2 support frequency-specific differentiated configuration, which can be realized by setting the **NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs** parameter to different values.

- Threshold for event A1
 - If the RFSP-specific coverage-based inter-frequency handover function is disabled, the threshold for event A1 cannot be customized for UEs. In this case, the threshold for event A1 is calculated as follows: $Q_{a1} + Q_{offset1} + Q_{offset2} + Q_{offset3}$.
 - If the RFSP-specific coverage-based inter-frequency handover function is enabled, the threshold for event A1 can be customized for UEs. In this

case, the threshold for event A1 is calculated as follows: $Q_{a1} + Q_{offset1} + Q_{offset2} + Q_{offset3} + Q_{offset4}$.

Table 4-10 Variables in the formulas for calculating the threshold for event A1

Variable	Parameter Setting	Description
Q_{a1}	NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld	This parameter specifies the RSRP threshold for event A1 related to coverage-based inter-frequency handover.
$Q_{offset1}$	gNBOperatorQciParam.InterFreqA1RsrpThldOfs	This parameter specifies the offset used to calculate the operator-specific RSRP threshold for inter-frequency event A1.
$Q_{offset2}$	NRCellMeasConfig.SulCovHoRsrpThldOffset	This parameter specifies the offset used to calculate the RSRP threshold for event A1 or A2 related to coverage-based inter-frequency handovers. This parameter takes effect only for SUL UEs.
$Q_{offset3}$	NRCellFreqRelation.InterFreqHoA1A2A51FreqOfs	This parameter specifies the frequency-specific offset to the RSRP thresholds for events A1 and A2 and RSRP threshold 1 for event A5 related to coverage-based inter-frequency handovers.
$Q_{offset4}$	gNBRfspConfig.InterFreqA1A2A5RsrpThd1Ofs	This parameter specifies the offset used to calculate the RSRP threshold 1 for events A1, A2, and A5 related to coverage-based inter-frequency handovers of UEs with the specified RFSP index.

- Threshold for event A2
 - If the RFSP-specific coverage-based inter-frequency handover function is disabled, the threshold for event A2 cannot be customized for UEs. In this case, the threshold for event A2 is calculated as follows: $Q_{a2} + Q'_{offset1} + Q_{offset2} + Q_{offset3}$.

- If the RFSP-specific coverage-based inter-frequency handover function is enabled, the threshold for event A2 can be customized for UEs. In this case, the threshold for event A2 is calculated as follows: $Q_{a2} + Q'_{offset1} + Q_{offset2} + Q_{offset3} + Q_{offset4}$.

Table 4-11 Variables in the formulas for calculating the threshold for event A2

Variable	Parameter Setting	Description
Q_{a2}	NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld	This parameter specifies the RSRP threshold for event A2 related to coverage-based inter-frequency handover.
$Q'_{offset1}$	gNBOperatorQciParam.InterFreqA2RsrpThldOfs	This parameter specifies the offset used to calculate the operator-specific RSRP threshold for inter-frequency event A2.
For details about the definitions of $Q_{offset2}$, $Q_{offset3}$, and $Q_{offset4}$, see Table 4-10 .		

- Threshold 1 for event A5
 - If the RFSP-specific coverage-based inter-frequency handover function is disabled, threshold 1 for event A5 cannot be customized for UEs. In this case, threshold 1 for event A5 is calculated as follows: $Q_{a5-1} + Q_{offset3}$.
 - If the RFSP-specific coverage-based inter-frequency handover function is enabled, threshold 1 for event A5 can be customized for UEs. In this case, threshold 1 for event A5 is calculated as follows: $Q_{a5-1} + Q_{offset3} + Q_{offset4}$.

Table 4-12 Variables in the formulas for calculating threshold 1 for event A5

Variable	Parameter Setting	Description
Q_{a5-1}	NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld1	This parameter specifies threshold 1, expressed as the RSRP in the serving cell, for event A5 related to coverage-based inter-frequency handovers.
For details about the definitions of $Q_{offset3}$ and $Q_{offset4}$, see Table 4-10 .		

- Threshold 2 for event A5
 - If the RFSP-specific coverage-based inter-frequency handover function is disabled, threshold 2 for event A5 cannot be customized for UEs. In this case, threshold 2 for event A5 is calculated as follows: $Q_{a5-2} + Q'_{offset1}$.

- If the RFSP-specific coverage-based inter-frequency handover function is enabled, threshold 2 for event A5 can be customized for UEs. In this case, threshold 2 for event A5 is calculated as follows: $Q_{a5-2} + Q''_{offset1} + Q'_{offset2}$.

Table 4-13 Variables in the formulas for calculating threshold 2 for event A5

Variable	Parameter Setting	Description
Q_{a5-2}	NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2	This parameter specifies threshold 2, expressed as the RSRP in a neighboring cell, for event A5 related to coverage-based inter-frequency handovers.
$Q''_{offset1}$	gNBOperatorQciParameter.InterFreqA5RsrpThld2Ofs	This parameter specifies the offset of the operator-specific RSRP threshold 2 for inter-frequency event A5 relative to the NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 parameter.
$Q'_{offset2}$	gNBRfspConfig.InterFreqA5RsrpThld2Ofs	This parameter specifies the offset used to calculate the RSRP threshold 2 for event A5 related to coverage-based inter-frequency handovers of UEs with the specified RFSP index.

4.1.2.2.3 Flexible Configuration in Coverage-based Inter-RAT Handover Scenarios

In scenarios of RFSP-specific coverage-based inter-RAT handovers between E-UTRAN and NG-RAN, if the RFSP-specific coverage-based inter-RAT handover function is enabled, thresholds for events A1 and A2 and threshold 1 for event B2 can be customized for UEs based on RFSPs. The RFSP-specific coverage-based inter-RAT handover function is controlled by the **NR_TO_EUTRAN_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

The gNodeB selects the thresholds for inter-RAT events A1 and A2, and threshold 1 for inter-RAT event B2 based on whether the RFSP-specific coverage-based inter-RAT handover function is enabled.

- Threshold for inter-RAT event A1
 - If the RFSP-specific coverage-based inter-RAT handover function is disabled, the threshold for inter-RAT event A1 cannot be customized for

UEs. In this case, the threshold for inter-RAT event A1 is calculated as follows: $Q_{\text{inter-a1}} + Q_{\text{inter-offset1}} + Q_{\text{inter-offset2}}$.

- If the RFSP-specific coverage-based inter-RAT handover function is enabled, the threshold for inter-RAT event A1 can be customized for UEs. In this case, the threshold for inter-RAT event A1 is calculated as follows: $Q_{\text{inter-a1}} + Q_{\text{inter-offset1}} + Q_{\text{inter-offset2}} + Q_{\text{inter-offset4}}$.

Table 4-14 Variables in the formulas for calculating the threshold for inter-RAT event A1

Variable	Parameter Setting	Description
$Q_{\text{inter-a1}}$	NRCellInterRHoMeaGrp.InterRatHoA1RsrpThld	This parameter specifies the RSRP threshold for event A1 related to inter-RAT handover.
$Q_{\text{inter-offset1}}$	gNBOperatorQciParam.EutranA1RsrpThldOffset	This parameter specifies the offset of the RSRP threshold for event A1 related to handover to E-UTRAN relative to NRCellInterRHoMeaGrp.InterRatHoA1RsrpThld .
$Q_{\text{inter-offset2}}$	NRCellMeasConfig.SulCovHoRsrpThldOffset	This parameter specifies the offset used to calculate the RSRP threshold for event A1 or A2 related to coverage-based inter-frequency handover. This parameter takes effect only for SUL UEs.
$Q_{\text{inter-offset4}}$	gNBRfspConfig.InterRatA1A2B2RsrpThldOfs	This parameter specifies the offset used to calculate the following thresholds related to coverage-based inter-RAT handovers of UEs with the specified RFSP index: RSRP threshold for event A1, RSRP threshold for event A2, and RSRP threshold 1 for event B2.

- Threshold for inter-RAT event A2
 - If the RFSP-specific coverage-based inter-RAT handover function is disabled, the threshold for inter-RAT event A2 cannot be customized for UEs. In this case, the threshold for inter-RAT event A2 is calculated as follows: $Q_{\text{inter-a2}} + Q_{\text{inter-offset1}} + Q_{\text{inter-offset2}}$.
 - If the RFSP-specific coverage-based inter-RAT handover function is enabled, the threshold for inter-RAT event A2 can be customized for UEs.

In this case, the threshold for inter-RAT event A2 is calculated as follows:

$$Q_{\text{inter-a2}} + Q'_{\text{inter-offset1}} + Q_{\text{inter-offset2}} + Q_{\text{inter-offset4}}$$

Table 4-15 Variables in the formulas for calculating the threshold for inter-RAT event A2

Variable	Parameter Setting	Description
$Q_{\text{inter-a2}}$	NRCellInterRHoMeaGrp.InterRatHoA2RsrpThld	This parameter specifies the RSRP threshold for event A2 related to inter-RAT handover.
$Q'_{\text{inter-offset1}}$	gNBOperatorQciParam.EutranA2B2RsrpThldOffset	For event A2, this parameter specifies the offset of the RSRP threshold for event A2 related to handover to E-UTRAN relative to the RSRP threshold for event A2 related to inter-RAT handover. For event B2, this parameter specifies the offset of the RSRP threshold 1 for event B2 related to handover to E-UTRAN relative to the RSRP threshold 1 for event B2 related to coverage-based handover to E-UTRAN.
For details about the definitions of $Q_{\text{inter-offset2}}$ and $Q_{\text{inter-offset4}}$, see Table 4-14 .		

- Threshold 1 for event B2
 - If the RFSP-specific coverage-based inter-RAT handover function is disabled, threshold 1 for event B2 cannot be customized for UEs. In this case, threshold 1 for event B2 is calculated as follows: $Q_{\text{inter-b2-1}} + Q'_{\text{inter-offset1}}$.
 - If the RFSP-specific coverage-based inter-RAT handover function is enabled, threshold 1 for event B2 can be customized for UEs. In this case, threshold 1 for event B2 is calculated as follows: $Q_{\text{inter-b2-1}} + Q'_{\text{inter-offset1}} + Q_{\text{inter-offset4}}$.

Table 4-16 Variables in the formulas for calculating threshold 1 for event B2

Variable	Parameter Setting	Description
$Q_{\text{inter-b2-1}}$	NRCellInterRHoMeaGrp.CovHoToEutranB2RsrpThld1	This parameter specifies the RSRP threshold 1 for event B2 related to coverage-based handover to E-UTRAN.

Variable	Parameter Setting	Description
For details about the definition of $Q'_{\text{inter-offset1}}$, see Table 4-15 .		
For the definition of $Q_{\text{inter-offset4}}$, see Table 4-14 .		

NOTE

The preceding thresholds require that all of the following conditions apply:

- If the calculated value of any of the preceding thresholds is greater than –31 dBm, the threshold that takes effect is –30 dBm.
- If the calculated value of any of the preceding thresholds is less than –156 dBm, the threshold that takes effect is –156 dBm.

4.1.2.2.4 Flexible Configuration in Frequency-Priority-based Inter-Frequency Handover Scenarios

The following two flexible configuration methods are available:

- Method 1: Flexible configuration of the handover event triggering mode. For details, see [Handover Event Triggering Mode](#).
- Method 2: Flexible configuration of traffic priorities, frequency lists, and handover event thresholds. For details, see [Traffic Priorities, Frequency Lists, and Handover Event Thresholds](#).

Handover Event Triggering Mode

In intra-RAT frequency-priority-based inter-frequency handover scenarios, the handover event triggering mode can be customized for UEs based on RFSPs. The general working principles are as follows: On the gNodeB, differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** (*gNBRfspConfig.RfspFreqPriHoTrigMode*) are configured for UEs with different RFSP indexes. In this way, differentiated triggering modes of frequency-priority-based inter-frequency handovers can take effect for different types of UEs.

Specifically:

1. The gNodeB obtains the RFSP indexes of UEs from the core network.
2. On the gNodeB, differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** are configured for UEs with different RFSP indexes. Then, the gNodeB determines the handover event triggering modes for UEs with specific RFSP indexes according to the values of this parameter.
 - a. Differentiated values of **Freq Priority HO Trigger Mode Based on RFSP** are configured for UEs with different RFSP indexes.

Parameter Name	Parameter ID	Description
Operator ID	<i>gNBRfspConfig.OperatorId</i>	This parameter specifies the operator of a UE.

Parameter Name	Parameter ID	Description
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.
Freq Priority HO Trigger Mode Based on RFSP	gNBRfspConfig.RfspFreqPriHoTriggerMode	This parameter specifies the triggering mode of frequency-priority-based handover for UEs with a specific RFSP index.

- b. The gNodeB determines the triggering mode of frequency-priority-based handover for UEs with a specific RFSP index.
- If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **A1_BASED** for an RFSP index, event A1 related to frequency-priority-based inter-frequency handover is triggered for UEs with this RFSP index.
 - If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **A2_BASED** for an RFSP index, event A2 related to frequency-priority-based inter-frequency handover is triggered for UEs with this RFSP index.
 - If the corresponding **Freq Priority HO Trigger Mode Based on RFSP** is set to **NULL** for an RFSP index, the **NRCellMobilityConfig.FreqPriHoTriggerMode** parameter is used to specify the frequency-priority-based inter-frequency handover event type for UEs with this RFSP index.

Traffic Priorities, Frequency Lists, and Handover Event Thresholds

In intra-RAT frequency-priority-based inter-frequency handover scenarios, if the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, the traffic priorities, frequency lists, thresholds for events A1, A2, and A4, and threshold 2 for event A5 can be customized for UEs based on RFSPs. The RFSP-specific frequency-priority-based inter-frequency handover function is controlled by the **FREQ_PRIORITY_BASED_HO_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter.

- Traffic priorities and frequency lists

As listed in [Table 4-17](#), the gNodeB selects traffic priorities and frequency lists to be delivered to UEs based on whether the RFSP-specific frequency-priority-based inter-frequency handover function is enabled.

Table 4-17 Frequency list used for frequency-priority-based inter-frequency handovers

RFSP-specific Frequency-Priority-based Inter-Frequency Handover Function	Traffic Priority	Frequency List
Enabled	gNBFreqPriorityGroup.TrafficPriority^a	Frequencies that meet both of the following requirements: The value of gNBFreqPriorityGroup.TrafficPriority to which gNBRfspConfig.RfspIndex is bound is not 0 and the frequencies have higher traffic priorities than the serving frequency.
Disabled	NRCellFreqRelation.TrafficPriority	Frequencies with NRCellFreqRelation.TrafficPriority set to a non-zero value
a: In the RFSP-specific dedicated frequency list group to which the gNBRfspConfig.RfspIndex parameter is bound, there must be at least one non-serving frequency with gNBFreqPriorityGroup.TrafficPriority set to a non-zero value. Otherwise, the RFSP-specific frequency-priority-based inter-frequency handover function cannot take effect. If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled but does not take effect, cell-level frequency-priority-based inter-frequency handover takes effect for UEs (that is, the traffic priority specified by the NRCellFreqRelation.TrafficPriority parameter is used).		

- Thresholds for events A1, A2, and A4, and threshold 2 for event A5

The gNodeB selects thresholds for events A1, A2, and A4 and threshold 2 for event A5 based on whether the RFSP-specific frequency-priority-based inter-frequency handover function is enabled.

- Threshold for event A1
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, the threshold for event A1 cannot be customized for UEs. In this case, the threshold for event A1 equals the value of **NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld**.
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, the threshold for event A1 can be customized for UEs. In this case, the threshold for event A1 equals the sum of the values of **NRCellInterFHoMeaGrp.FreqPriInterFA1RsrpThld** and **gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs**.
- Threshold for event A2

- If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, the threshold for event A2 cannot be customized for UEs. In this case, the threshold for event A2 equals the value of **NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld**.
- If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, the threshold for event A2 can be customized for UEs. In this case, the threshold for event A2 equals the sum of the values of **NRCellInterFHoMeaGrp.FreqPriInterFA2RsrpThld** and **gNBRfspConfig.FreqPriHoA1A2RsrpThldOfs**.
- Threshold for event A4
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, the threshold for event A4 cannot be customized for UEs. In this case, the threshold for event A4 equals the value of **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld**.
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, the threshold for event A4 can be customized for UEs. In this case, the threshold for event A4 equals the sum of the values of **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld** and **gNBRfspConfig.FreqPriFHoA4RsrpThldOfs**.
- Threshold 2 for event A5
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is disabled, threshold 2 for event A5 cannot be customized for UEs. In this case, threshold 2 for event A5 equals the value of **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld**.
 - If the RFSP-specific frequency-priority-based inter-frequency handover function is enabled, threshold 2 for event A5 can be customized for UEs. In this case, threshold 2 for event A5 equals the sum of the values of **NRCellInterFHoMeaGrp.FreqPriInterFA4RsrpThld** and **gNBRfspConfig.FreqPriFHoA4RsrpThldOfs**.

4.1.2.2.5 Flexible Configuration in Service-based Inter-Frequency Handover Scenarios

In RFSP-specific service-based inter-frequency handover scenarios, the threshold for event A4 and threshold 2 for event A5 related to service-based inter-frequency handover can be customized for UEs based on RFSPs.

- Threshold for event A4 = **NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld** + **gNBRfspConfig.ServBasedInterFHoA4RsrpOfs**
- Threshold 2 for event A5 = **NRCellInterFHoMeaGrp.SrvInterFreqA4RsrpThld** + **gNBRfspConfig.ServBasedInterFHoA4RsrpOfs**

NOTE

Threshold 2 for event A5 related to service-based inter-frequency handover, which is customized for UEs based on RFSPs, reuses the parameters of the threshold for event A4.

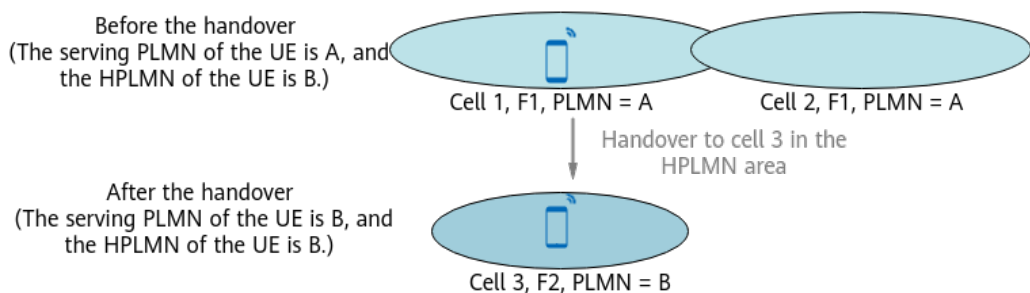
4.1.2.3 Policy of RFSP-based Inter-Frequency Handover Back to the HPLMN

When the serving PLMN where a UE roams and its HPLMN have overlapped coverage, the policy of RFSP-based inter-frequency handover back to the HPLMN can be used to control whether the roaming UE uses the RFSP-specific dedicated frequency list during cell handovers so that it can be quickly handed over back to a cell on the HPLMN frequency, as shown in Figure 4-3. It is recommended that an operator enable this function when the home network coverage area of a roaming UE overlaps with the local network coverage area.

NOTE

- HPLMN: indicates the subscribed home operator of a UE.
- Serving PLMN: indicates an operator that provides services for a UE. If a UE camps on its home network, the serving PLMN is the HPLMN of the UE.
- Roaming UE: If a UE leaves the network coverage area of its home operator and accesses another network (with which the UE needs to sign a roaming agreement), the UE is referred to as a roaming UE on that network.

Figure 4-3 RFSP-based inter-frequency handover back to the HPLMN



RFSP Indexes for Roaming UEs

A gNodeB cannot directly identify roaming UEs, but can identify their RFSP indexes. Therefore, operators need to plan a specific range of RFSP indexes for roaming UEs on the core network. In the current version, the gNodeB does not support an RFSP index change. Therefore, RFSP indexes must be jointly planned among operators to ensure that the RFSP index ranges used by different operators do not overlap.

For example, the RFSP index range is 1 to 50 for UEs subscribed to operator A, and 51 to 100 for UEs subscribed to operator B. When UEs of operator B access or are handed over to operator A's network, operator A's core network delivers RFSP indexes in the range from 51 to 100 to the gNodeB. Then, the gNodeB can identify UEs with the RFSP indexes in the range from 51 to 100 on operator A's network as roaming UEs.

On the gNodeB side, the **gNBRfspHplmn** MO is used to configure the RFSP indexes and HPLMNs of roaming UEs. Table 4-18 lists the related parameters.

Table 4-18 Parameters for configuring the RFSP indexes and HPLMNs of roaming UEs

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspHplmn.OperatorId	These parameters specify the RFSP index range of roaming UEs served by an operator.
Start RFSP Index	gNBRfspHplmn.StartRfspIndex	
End RFSP Index	gNBRfspHplmn.EndRfspIndex	
Home Mobile Country Code	gNBRfspHplmn.HomeMcc	These parameters specify the HPLMN of roaming UEs.
Home Mobile Network Code	gNBRfspHplmn.HomeMnc	

RFSP-specific Dedicated Frequency List of Roaming UEs

The following describes how the RFSP-specific dedicated frequency lists for roaming UEs are configured and delivered to UEs on the gNodeB side as well as how an RFSP-specific dedicated frequency list is used for handovers:

1. Configure frequency list groups on the gNodeB side.

The **gNBFreqPriorityGroup** MO is used to configure the frequency list group required during cell handovers of roaming UEs. [Table 4-19](#) lists major parameters in this MO.

Table 4-19 gNodeB frequency priority group

Parameter Name	Parameter ID	Description
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	This parameter specifies the ID of a gNodeB frequency priority group.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	This parameter specifies the index of a frequency in the gNodeB frequency priority group.
RAT Type	gNBFreqPriorityGroup.RatType	This parameter specifies the RAT corresponding to the configured frequency.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	This parameter specifies the frequency-domain position of the SSB on an NR frequency.

Parameter Name	Parameter ID	Description
Downlink EARFCN	gNBFreqPriorityGroup.DLEarfcn	This parameter specifies the downlink frequency of a neighboring E-UTRAN cell.
Traffic Priority	gNBFreqPriorityGroup.TrafficPriority	This parameter specifies the traffic priority of a non-serving NR frequency.
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag	This parameter needs to be set to TRUE for a frequency belonging to the HPLMN of the roaming UE.

Table 4-20 lists configuration examples of gNodeB frequency list groups.

Table 4-20 Configuration examples of gNodeB frequency list groups

gNodeB Frequency Priority Group ID	Frequency Index	RAT Type	SSB Frequency Position	Downlink EARFCN	Roaming UE HPLMN Frequency Flag
0	0	NR	F1	N/A	FALSE
0	1	NR	F2	N/A	TRUE

Note: F1 and F2 in the table exemplify different frequencies.

- The gNodeB obtains the RFSP indexes of UEs from the core network.
- Based on the obtained RFSP indexes, bind the RFSP indexes to the configured gNodeB frequency list groups on the gNodeB to generate the RFSP-specific dedicated frequency lists.

The **gNBRfspConfig** MO is used for the binding. **Table 4-21** lists the related parameters.

Table 4-21 Parameters for binding an RFSP index to a gNodeB frequency list group

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.

Parameter Name	Parameter ID	Description
gNodeB Frequency Priority Group ID	gNBRfspConfig.gNBFreqPriorityGroupId	This parameter specifies the ID of the gNodeB frequency list group bound to the RFSP index. This parameter has a mapping relationship with the gNBFreqPriorityGroup.gNBFreqPriorityGroupId parameter. If this parameter is set to 255, the gNodeB frequency list group bound to the RFSP index is invalid.

NOTE

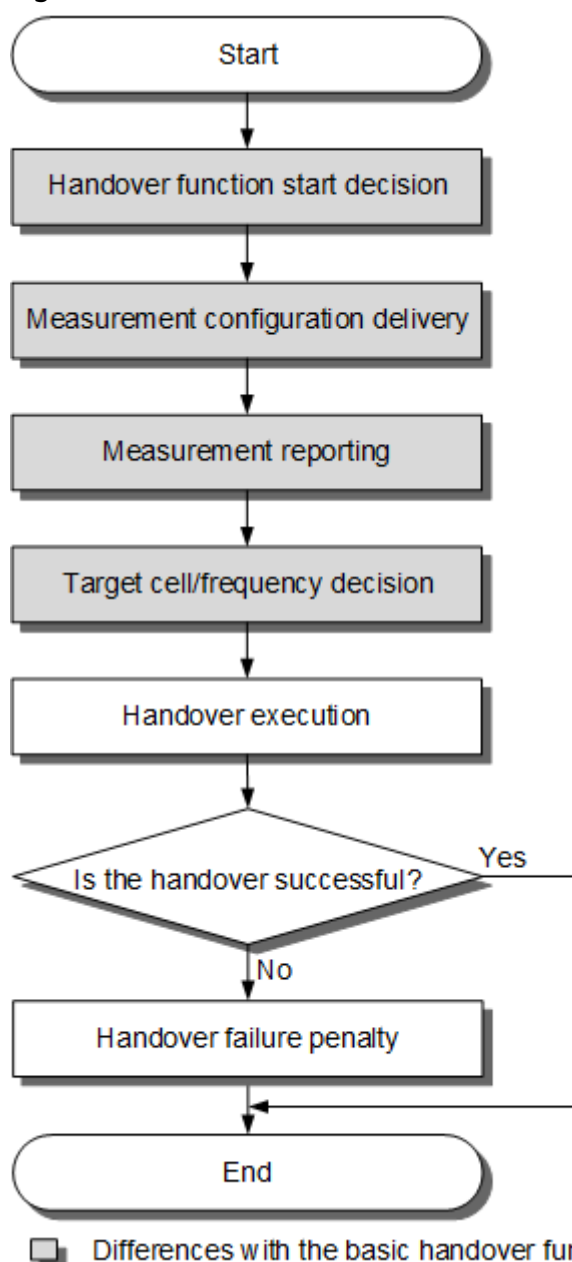
In RedCap/eRedCap scenarios, the gNodeB frequency list group ID bound to the RFSP index is specified by the **gNBRfspConfig.RedCapGnbFreqPriGroupId** parameter. For details, see the section "Frequency Selection for RedCap" in *RedCap*.

4. The gNodeB identifies roaming UEs based on the obtained RFSP indexes.
The gNodeB identifies roaming UEs. For detailed principles, see [RFSP Indexes for Roaming UEs](#). However, the gNodeB does not identify a UE as a roaming UE in the following scenarios:
 - The gNodeB fails to obtain the RFSP index of the UE from the core network.
 - The gNodeB has obtained the RFSP index of the UE from the core network. Although the RFSP index is within the RFSP index range of roaming UEs, the RFSP index is not specified by any **gNBRfspConfig.RfspIndex** parameter.
 - The gNodeB has obtained the RFSP index of the UE from the core network. Although the RFSP index is within the RFSP index range of roaming UEs, the gNodeB frequency list group bound to the RFSP index does not contain the frequency whose **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** is **TRUE**.
5. The gNodeB determines whether RFSP-based inter-frequency handover back to the HPLMN takes effect for the roaming UE corresponding to the RFSP index.
 - If the RFSP index is bound to a valid gNodeB frequency list group, RFSP-based inter-frequency handover back to the HPLMN takes effect for the roaming UE of the RFSP index. In this case, the process proceeds to [6](#).
 - If the RFSP index is bound to an invalid gNodeB frequency list group, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the roaming UE. In this case, the process ends.
6. The gNodeB delivers the RFSP-specific dedicated frequency list to the roaming UE through the RRCReconfiguration message.

7. The roaming UE obtains the RFSP-specific dedicated frequency list and checks configurations related to its RFSP index in the RFSP-specific dedicated frequency list to determine whether RFSP-based inter-frequency handover back to the HPLMN takes effect.
 - If the RFSP index of the roaming UE has a corresponding **gNBRRfspConfig.RfspIndex**, the UE uses the gNodeB frequency list group bound to the RFSP index for RFSP-based inter-frequency handover back to the HPLMN.
 - If the RFSP index of the roaming UE does not have a corresponding **gNBRRfspConfig.RfspIndex**, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the UE.

Procedure of RFSP-based Inter-Frequency Handover Back to the HPLMN

Figure 4-4 shows the procedure of RFSP-based inter-frequency handover back to the HPLMN. This section describes only the differences between this procedure and the basic cell handover functions. For details about the basic cell handover functions, see *SA Mobility Management in Connected Mode*.

Figure 4-4 Procedure of RFSP-based inter-frequency handover back to the HPLMN

- Handover function start decision

RFSP-based inter-frequency handover back to the HPLMN supports only the measurement-based mode, rather than the blind mode. This function is enabled when all of the following conditions are met:

- The **BACK_TO_HPLMN_INTER_FREQ_HO_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter is selected.
- The **INTRA_RAT_HO_WITH_HPLMN_SW** option of the **gNodeBAlgo.HplmnAlgoSwitch** parameter is selected.
- The **INTER_FREQ_LIST_CTRL_SW** option of the **NRCellAlgoSwitch.RfspAlgoSwitch** parameter is selected.

- In the frequency list group bound to the RFSP index of the roaming UE, there is a frequency with **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** set to **TRUE**, and this frequency must be a neighboring frequency of the serving frequency.
- Measurement configuration delivery
After the RFSP-based inter-frequency handover back to the HPLMN function is enabled, the gNodeB selects the frequencies with **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** being **TRUE** in the frequency list group bound to the RFSP index of the roaming UE and delivers the measurement configuration to the UE.
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A4**, the gNodeB delivers measurement configurations for event A4 related to RFSP-based inter-frequency handover back to the HPLMN.
 - If the **NRCellMobilityConfig.InterFreqMeasEventConfig** parameter is set to **EVENT_A5**, the gNodeB delivers measurement configurations for event A5 related to RFSP-based inter-frequency handover back to the HPLMN.

In the preceding information:

- Event A4 related to RFSP-based inter-frequency handover back to the HPLMN: Indicates that the signal quality of a neighboring cell exceeds a specific threshold (controlled by the **gNBRfspConfig.BackToHplmnHoA5RsrpThld2** parameter).
- Event A5 related to RFSP-based inter-frequency handover back to the HPLMN: Indicates that the signal quality of the serving cell drops below threshold 1 (controlled by the **gNBRfspConfig.BackToHplmnHoA5RsrpThld1** parameter) and the signal quality of a neighboring cell exceeds threshold 2 (controlled by the **gNBRfspConfig.BackToHplmnHoA5RsrpThld2** parameter).

In SA networking, threshold offsets can be used to enable differentiated configurations of thresholds for events A4 and A5 for voice service UEs (that is, UEs performing VoNR services). The following lists the thresholds for events A4 and A5 for voice service UEs:

- Threshold 1 for event A5: **gNBRfspConfig.BackToHplmnHoA5RsrpThld1 + gNBRfspConfig.VoBackHplmnA5RsrpThld1Ofs**
- Threshold 2 for event A5 and threshold for event A4: **gNBRfspConfig.BackToHplmnHoA5RsrpThld2 + gNBRfspConfig.VoBackHplmnA5RsrpThld2Ofs**

If the event A4 or A5 measurement for RFSP-based inter-frequency handover back to the HPLMN has been performed for 6s but no neighboring cell that meets the requirements is detected, the gNodeB deletes the corresponding measurement configurations and instructs the UE to stop the event A4 or A5 measurement. In addition, the gNodeB periodically delivers event A4 or A5 measurement configurations to UEs that experience measurement failures. The period can be specified by the **gNBRfspConfig.BackToHplmnHoRetryPeriod** parameter. If this parameter is set to **255**, the gNodeB does not deliver event A4 or A5 measurement configurations again to UEs that experience measurement failures.

 NOTE

To prevent UEs of the current operator from measuring frequencies of roaming UEs' HPLMNs, set the **NRCellFreqRelation.InterFreqPrdcMrType** parameter (indicating the inter-frequency periodic MR measurement type) to **FORBID_MEAS** for the frequency.

The settings of inter-frequency periodic MR measurement type do not apply to frequencies whose **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** is set to **TRUE** in the frequency list group bound to the RFSP indexes of roaming UEs.

- Measurement reporting

The UE performs measurements based on the event measurement configurations and reports the measurement result.

- Target cell or frequency decision

When the gNodeB receives the measurement report on the event A4 or A5 measurement for RFSP-based inter-frequency handover back to the HPLMN, it obtains the frequency information from the measurement report and determines the target cell as follows:

- For a non-multi-operator sharing cell in the measurement report:
 - If the PLMN of the cell is the HPLMN of the roaming UE, the gNodeB determines that the cell is the target cell and performs a handover. In addition, the serving PLMN of the UE is set to the HPLMN of the UE.
 - If the PLMN of the cell is not the HPLMN of the roaming UE, the handover is not performed.
- For a multi-operator sharing cell in the measurement report:
 - If PLMNs of the cell include the HPLMN of the roaming UE, the gNodeB determines that the cell is the target cell and performs a handover. In addition, the serving PLMN of the UE is set to the HPLMN of the UE.
 - If PLMNs of the cell do not include the HPLMN of the roaming UE, the handover is not performed.

4.1.2.4 RFSP-based Unnecessary Handover Policy

If the unnecessary handover function switch is turned on, the gNodeB can determine whether the unnecessary handover function takes effect for specific UEs based on RFSP indexes through the RFSP-specific unnecessary handover policy.

 NOTE

- An unnecessary handover is performed depending on network deployment policies. The serving cell can still provide services for the UEs even if the handover fails. This document describes only the content exclusive to the RFSP-based unnecessary handover policy compared with basic unnecessary handover functions.
- For details about classification of handover functions (necessary and unnecessary handovers), see *SA Mobility Management in Connected Mode*.

The general principle of the RFSP-based unnecessary handover policy is as follows: On the gNodeB, the **Unnecessary Handover Suppression Switch** (specified by the **UNNECESSARY_HO_SUPPRESS_SW** option of the

gNB RfspConfig.RfspAlgoSwitch parameter) is configured differentially for UEs with different RFSP indexes. In this way, differentiated unnecessary handover policies take effect for different UEs. In the current version, this function can be enabled only in low frequency bands.

Specifically:

1. The gNodeB obtains the RFSP indexes of UEs from the core network.
2. Based on the obtained RFSP indexes, the gNodeB configures the **Unnecessary Handover Suppression Switch** differentially for RFSP indexes. Then, the gNodeB determines whether the unnecessary handover function takes effect for the UEs with specified RFSP indexes based on the switch status.
 - a. The **Unnecessary Handover Suppression Switch** is configured differentially for RFSP indexes.

Parameter Name	Parameter ID	Description
Operator ID	gNB RfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNB RfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.
Unnecessary Handover Suppression Switch	UNNECESSARY_HO_SUPPRESS_SW option of the gNB RfspConfig.RfspAlgoSwitch parameter	This option specifies whether to suppress unnecessary handovers for UEs with the specified RFSP.

- b. The gNodeB determines whether the unnecessary handover function takes effect for UEs with specified RFSP indexes.
 - If the RFSP indexes of UEs have corresponding **gNB RfspConfig.RfspIndex** values, the UEs determine whether the unnecessary handover function takes effect based on the status of the **Unnecessary Handover Suppression Switch** corresponding to the RFSP indexes.
If the switch is turned on, the unnecessary handover function does not take effect. If the switch is turned off, the unnecessary handover function takes effect for UEs when the conditions for the function to take effect are met.
 - If the RFSP indexes of UEs do not have corresponding **gNB RfspConfig.RfspIndex** values, the configuration of RFSP index 0 is checked.
If the **gNB RfspConfig.RfspIndex** parameter is 0 for any index and the **Unnecessary Handover Suppression Switch** is turned on for the index, the unnecessary handover function does not take effect. Otherwise, this function can take effect for UEs if the conditions for the function to take effect are met.

4.2 Network Analysis

4.2.1 Benefits

This function helps operators customize mobility management policies for different types of UEs to improve user experience. The gains of some sub-functions are as follows:

Policy of RFSP-based inter-frequency handover back to the HPLMN: This function enables a roaming UE to return to its HPLMN as soon as possible if the coverage area of the HPLMN overlaps with that of the UE's serving network, improving user experience of roaming UEs.

RFSP-based unnecessary handover policy: Operators can customize unnecessary handover policies for different types of UEs to improve service experience of these UEs.

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

After the RFSP-based NR inter-frequency handover policy or RFSP-based NG-RAN-to-E-UTRAN handover policy is enabled, only specified frequencies or cells provide services for specific UEs. As a result, the **Service Call Drop Rate (CU)** or **Service Call Drop Rate (CU, Inactive)** may increase.

After the RFSP-based unnecessary handover policy takes effect, UEs with the specified RFSP index do not perform unnecessary handovers. As a result, the number of handovers decreases, and the handover success rate and uplink and downlink UE throughputs fluctuate.

The definitions of the indicators involved are described as follows:

Table 4-22 Definition of related indicators

Indicator	Formula
Number of outgoing inter-frequency intra-gNodeB handover executions	N.HO.InterFreq.IntragNB.ExecAttOut
Number of outgoing inter-frequency Xn-based handover executions	N.HO.InterFreq.Xn.IntergNB.ExecAttOut
Number of NG-based inter-gNodeB inter-frequency outgoing handover executions	N.HO.InterFreq.Ng.IntergNB.ExecAttOut

Indicator	Formula
Number of outgoing frequency-priority-based inter-frequency handover executions	N.HO.InterFreq.FreqPri.ExecAttOut
Number of outgoing service-based inter-frequency handover executions	N.HO.InterFreq.Service.ExecAttOut
Number of outgoing handover executions from NR to E-UTRAN	N.HO.InterRAT.N2E.ExecAttOut
Outgoing inter-frequency handover success rate	Inter-Frequency Handover Out Success Rate (CU)
NR-to-LTE outgoing handover success rate	Inter-RAT Handover Out Success Rate (NR to LTE) (CU)
Uplink UE throughput	User Uplink Average Throughput (DU)
Downlink UE throughput	User Downlink Average Throughput (DU)

Other sub-functions have no impact on the network.

4.3 Requirements

4.3.1 Licenses

There are no license requirements for basic functions.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Cell selection and reselection	None	<i>SA Mobility Management in Idle Mode</i>	The RFSP-based cell reselection policy requires the cell selection and cell reselection function to be enabled.
Low-frequency TDD High-frequency TDD	Coverage-based inter-frequency handover	COVERAGE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In coverage-based inter-frequency handover scenarios, the RFSP-based handover policy requires the basic coverage-based inter-frequency handover function to be enabled.
Low-frequency TDD High-frequency TDD	RFSP-based NR inter-frequency handover policy	INTER_FREQ_LIST_CTRL_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter	<i>Flexible User Steering</i>	If RFSP-specific frequency-priority-based inter-frequency handover is enabled (by selecting the FREQ_PRIORITY_BASED_HO_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter), the RFSP-based NR inter-frequency handover policy must also be enabled.
Low-frequency TDD High-frequency TDD	RFSP-based NR inter-frequency handover policy	INTER_FREQ_LIST_CTRL_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter	<i>Flexible User Steering</i>	If the policy of RFSP-based inter-frequency handover back to the HPLMN is enabled (by selecting the BACK_TO_HPLMN_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter), the RFSP-based NR inter-frequency handover policy must also be enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Intra-RAT inter-PLMN handovers based on HPLMNs	INTRA_RAT_HO_WITH_HPLMN_SW option of the gNodeBAlgo.HplmnAlgoSwitch parameter	None	If the policy of RFSP-based inter-frequency handover back to the HPLMN is enabled (by selecting the BACK_TO_HPLMN_INTER_FREQ_HO_SW option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter), intra-RAT inter-PLMN handovers based on HPLMNs must also be enabled.
Low-frequency TDD High-frequency TDD	Operator-specific-priority-based inter-frequency handover	OP_DED_PRI_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In operator-specific-priority-based inter-frequency handover scenarios, the basic operator-specific-priority-based inter-frequency handover function must be enabled before the RFSP-based handover policy is enabled.
Low-frequency TDD High-frequency TDD	Service-based inter-frequency handover	SERVICE_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In service-based inter-frequency handover scenarios, the basic service-based inter-frequency handover function must be enabled before the RFSP-based handover policy is enabled.
Low-frequency TDD	Uplink-interference-based inter-frequency handover	UL_INTRF_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In uplink-interference-based inter-frequency handover scenarios, the basic uplink-interference-based inter-frequency handover function must be enabled before the RFSP-based handover policy is enabled.
Low-frequency TDD	Voice-quality-based inter-frequency handover	QLTY_BASED_INTER_FREQ_HO_SW option of the gNBQciAlgoParameterGrp.QciAlgoSwitch parameter	<i>VoNR</i>	In voice-quality-based inter-frequency handover scenarios, the basic voice-quality-based inter-frequency handover function must be enabled before the RFSP-based handover policy is enabled.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	SSB SINR-based inter-frequency handover	SSB_SINR_BASSED_HO option of the NRCellAlgoSwitch . InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In SSB SINR-based inter-frequency handover scenarios, the RFSP-based handover policy requires the basic SSB SINR-based inter-frequency handover function to be enabled.
Low-frequency TDD High-frequency TDD	Inter-RAT mobility from NG-RAN to E-UTRAN	MOBILITY_TO_EUTRAN_SWITCH option of the NRCellAlgoSwitch . InterRatServiceMobilitySwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i> <i>Interoperability Between E-UTRAN and NG-RAN in Idle Mode</i>	- In scenarios of cell reselection from NG-RAN to E-UTRAN, the basic inter-RAT mobility from NG-RAN to E-UTRAN function must be enabled before the RFSP-based NG-RAN-to-E-UTRAN cell reselection policy is enabled. - In scenarios of coverage-based handover from NG-RAN to E-UTRAN, the basic inter-RAT mobility from NG-RAN to E-UTRAN function must be enabled before the RFSP-based NG-RAN-to-E-UTRAN handover policy is enabled.
Low-frequency TDD High-frequency TDD	EPS Fallback	EPS_FB_SWITCH option of the NRCellAlgoSwitch . VoiceStrategySwitch parameter	<i>Interoperability Between E-UTRAN and NG-RAN in Connected Mode</i>	In voice service scenarios, the RFSP-based handover policy requires the basic EPS fallback function to be enabled.

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Low-frequency deprioritisation-based idle mode MLB	NRCellMlbPolicy set to DECREASE_PRIORITY and the INTER_FREQ_IDLE_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter selected	<i>Mobility Load Balancing in Idle Mode</i>	Low-frequency deprioritisation-based idle mode MLB does not take effect for UEs configured with RFSP-specific dedicated cell-reselection priorities.
High-frequency TDD	High-frequency UE-number-based idle mode MLB	INTER_FREQ_IDLE_MLB_SW option of the NRCellAlgoSwitch.MlbAlgoSwitch parameter	<i>Mobility Load Balancing in Idle Mode</i>	High-frequency UE-number-based idle mode MLB does not take effect for UEs configured with RFSP-specific dedicated cell-reselection priorities.
Low-frequency TDD High-frequency TDD	Operator-specific-priority-based inter-frequency handover	OP_DED_PRIORITY_BASED_HO option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	When both the policy of RFSP-based inter-frequency handover back to the HPLMN and the operator-specific-priority-based inter-frequency handover function are enabled, only the policy of RFSP-based inter-frequency handover back to the HPLMN can be triggered for a roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	In the decision-making phase for RFSP-based inter-frequency handover back to the HPLMN, if (1) voice-capability-based inter-frequency handover back to the HPLMN is enabled for the home operator of a VoNR-incapable roaming UE, and (2) the BACK_TO_HPLMN_INTER_FREQ_HO_SWITCH option of the NRCellAlgoSwitch.InterFreqHoSwitch parameter is selected for the serving cell, then: - When the gNBFreqPriorityGroup.RoamingUEHplmnInterFHoFlag parameter is set to TRUE for the target cell's frequency in the frequency group associated with the roaming UE, the following mechanism is applied: - In the case that the roaming UE supports handovers, the gNodeB instructs the roaming UE to immediately perform a

RAT	Function Name	Function Switch	Reference	Description
				handover to the target cell. - In the case that the roaming UE does not support handovers, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the roaming UE. • When the gNBFreqPriorityGroup.RoamingUeHplmnInterFHoFlag parameter is set to FALSE for the target cell's frequency in the frequency group associated with the roaming UE, RFSP-based inter-frequency handover back to the HPLMN does not take effect for the roaming UE.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-frequency handover policy for voice service-capable UEs	VOICE_INTE R_FREQ_HO _POLICY_SW option of the NRCellFreqR elation.Aggr egationAttri bute parameter	<i>SA Mobility Management in Connected Mode</i>	If the "inter-frequency handover policy for voice-service-capable UEs" is enabled for a visited network's frequency in a home network's cell, the home network's gNodeB will not send an RRCRelease message containing the frequency to a VoNR-incapable UE of the home operator in roaming scenarios, and the UE will not reselect this frequency based on RFSP-specific dedicated priorities. For details about how to identify UEs that do not support VoNR on the home network in roaming scenarios and the scenarios where turning on the function switch of the inter-frequency handover policy for voice service-capable UEs or voice-capability-based inter-frequency handover back to the HPLMN is recommended, see <i>SA Mobility Management in Connected Mode</i>.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgorithmSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If the "voice-capability-based inter-frequency handover back to the HPLMN" function is enabled for the home network on the visited network's gNodeB, the gNodeB will send an RRCRelease message to a VoNR-incapable UE of the home operator in roaming scenarios. The message contains the frequencies for which gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag is set to TRUE in the frequency list group bound to the roaming UE on the base station side. In this case, the UE can reselect this frequency based on RFSP-specific dedicated priorities. For details about how to identify UEs that do not support VoNR on the home network in roaming scenarios and the scenarios where turning on the function switch of the inter-frequency handover policy for voice service-capable UEs or voice-capability-based inter-frequency handover back to the HPLMN is

RAT	Function Name	Function Switch	Reference	Description
				recommended, see <i>SA Mobility Management in Connected Mode</i> .

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite support the function only in low-frequency scenarios.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

RFSP-based unnecessary handover policy:

All NR-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

Other sub-functions:

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- UEs
UEs must comply with 3GPP Release 15 or later.
The RFSP-based handover policy does not take effect for UEs that are performing emergency calls.

- Core network
 - The core network must comply with 3GPP Release 15 or later, and the AMF must support RFSP configuration.
 - Inter-RAT handover requires the N26 interface to be configured between the mobility management entity (MME) and AMF.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-23 describes the parameters used for function activation.

Table 4-23 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	Set this parameter based on the network plan.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	Set this parameter based on the network plan.
RAT Type	gNBFreqPriorityGroup.RatType	• If an E-UTRAN frequency is to be configured, set this parameter to EUTRAN. • If an NR frequency is to be configured, set this parameter to NR.
Downlink EARFCN	gNBFreqPriorityGroup.DLEarfcn	Set this parameter based on the network plan.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	gNBFreqPriorityGroup.SsbDescMethod	Set this parameter to SSB_DESC_TYPE_GSCN .
Cell Reselection Priority	gNBFreqPriorityGroup.CellReseIPri	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPri	Set this parameter based on the network plan.
Traffic Priority	gNBFreqPriorityGroup.TrafficPriority	Set this parameter based on the network plan.
Roaming UE HPLMN Frequency Flag	gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag	Set this parameter based on the network plan.
Operator ID	gNBRfspConfig.OperatorId	Set this parameter based on the network plan.
Index to RAT/Frequency Selection Priority	gNBRfspConfig.RfspIndex	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	gNBRfspConfig.gNBFreqPriorityGroupId	Set this parameter based on the network plan.
Inter-Freq Measurement Event Configuration	NRCellMobilityConfig.InterFreqMeasEventConfig	Use the recommended value.

Parameter Name	Parameter ID	Setting Notes
RFSP Based Algorithm Switch	NRCellAlgoSwitch.RfspAlgoSwitch	- Select the INTER_FREQ_LIST_CTRL_SW option if the UE is expected to use the RFSP-based NR inter-frequency handover policy. Select the NR_TO_EUTRAN_FREQ_LIST_CTRL_SW option if the UE is expected to use the RFSP-based NG-RAN-to-E-UTRAN handover policy. - To enable RFSP-specific frequency-priority-based inter-frequency handover when INTER_FREQ_LIST_CTRL_SW is selected, select the FREQ_PRIORITY_BASED_HO_SW option. - To enable RFSP-based inter-frequency handover back to the HPLMN when INTER_FREQ_LIST_CTRL_SW is selected, select the BACK_TO_HPLMN_INTER_FREQ_HO_SW option.
HPLMN Algo Switch	gNodeBAlgo.HplmnAlgoSwitch	Select the INTRA_RAT_HO_WITH_HPLMN_SW option before enabling RFSP-based inter-frequency handover back to the HPLMN.
Start RFSP Index	gNBRfspHplmn.StartRfspIndex	Set this parameter based on the network plan.
End RFSP Index	gNBRfspHplmn.EndRfspIndex	Set this parameter based on the network plan.
Home Mobile Country Code	gNBRfspHplmn.HomeMcc	Set this parameter based on the network plan.
Home Mobile Network Code	gNBRfspHplmn.HomeMnc	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Back to HPLMN Handover A5 RSRP Threshold 1	gNBRfspConfig.BackToHplmnHoA5RsrpThld1	Use the recommended value.
Back to HPLMN Handover A5 RSRP Threshold 2	gNBRfspConfig.BackToHplmnHoA5RsrpThld2	Use the recommended value.
Back to HPLMN Handover Retry Period	gNBRfspConfig.BackToHplmnHoRetryPeriod	Use the recommended value.
RFSP Algorithm Switch	gNBRfspConfig.RfspAlgoSwitch	It is recommended that the UNNECESSARY_HO_SUPPRESS_SW option be deselected for an RFSP index if unnecessary handovers are required for UEs with the RFSP index. If unnecessary handovers are not required for UEs with a specific RFSP index, it is recommended that the UNNECESSARY_HO_SUPPRESS_SW option be selected for the RFSP index.

4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- RFSP-based cell reselection policy and basic RFSP-based handover policy TDD configuration examples

```
//Running the following base-station-level commands to configure frequency priority groups, NR
frequencies in the groups, cell reselection priorities, and traffic priorities of the NR frequencies
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, CellReselPri=5, CellReselSubPri=0DOT2,
TrafficPriority=1;
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6300, CellReselPri=3, CellReselSubPri=0DOT2,
TrafficPriority=2;
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=4, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8500, CellReselPri=5, CellReselSubPri=0DOT2,
TrafficPriority=2;
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=5, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=7800, CellReselPri=3, CellReselSubPri=0DOT2,
TrafficPriority=1;
//Running the following base-station-level commands to configure frequency priority groups, E-
UTRAN frequencies in the groups, and cell reselection priorities of the E-UTRAN frequencies
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DEarfcn=37900, CellReselPri=4, CellReselSubPri=0DOT2;
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1, RatType=EUTRAN,
DEarfcn=38100, CellReselPri=4, CellReselSubPri=0DOT2;
//Running the following base-station-level commands to bind RFSP indexes to the configured
frequency priority groups
ADD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=2, gNBFreqPriorityGroupId=0;
ADD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=3, gNBFreqPriorityGroupId=1;
//Running the following cell-level command to activate the RFSP-based NR inter-frequency handover
policy for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=INTER_FREQ_LIST_CTRL_SW-1;
//Running the following cell-level command to activate the RFSP-specific frequency-priority-based
inter-frequency handover function for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=FREQ_PRIORITY_BASED_HO_SW-1;
//Running the following cell-level command to activate the RFSP-based NG-RAN-to-E-UTRAN
handover policy for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=NR_TO_EUTRAN_FREQ_LIST_CTRL_SW-1;
```

- Policy of RFSP-based inter-frequency handover back to the HPLMN

TDD configuration examples

```
//Running the following base-station-level commands to configure frequency priority groups, NR
frequencies in the groups, and roaming UE HPLMN frequency flags
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=8000, RoamingUeHplmnFreqFlag=FALSE;
ADD GNBFRFQPRORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR,
SsbDescMethod=SSB_DESC_TYPE_GSCN, SsbFreqPos=6300, RoamingUeHplmnFreqFlag=TRUE;
//Configuring the type of measurement event for unnecessary inter-frequency handovers (event A5 as
an example)
MOD NRCELLMOBILITYCONFIG: NrCellId=0, InterFreqMeasEventConfig=EVENT_A5;
//Running the following base-station-level command to bind the RFSP index of roaming UEs to a
configured frequency priority group
ADD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=2, gNBFreqPriorityGroupId=0;
//Running the following base-station-level command to configure the HPLMN corresponding to the
RFSP index of roaming UEs
ADD GNBFRFSPHPLMN: OperatorId=0, StartRfspIndex=2, EndRfspIndex=2, HomeMcc="320",
HomeMnc="20";
//Running the following cell-level commands to activate the policy of RFSP-based inter-frequency
handover back to the HPLMN for the current cell (NrCellId = 0) (The INTER_FREQ_LIST_CTRL_SW
option of the NRCellAlgoSwitch.RfspAlgoSwitch parameter and the
INTRA_RAT_HO_WITH_HPLMN_SW option of the gNodeBAlgo.HplmnAlgoSwitch parameter need to
be selected before the activation.)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=INTER_FREQ_LIST_CTRL_SW-1;
MOD GNODEBALGO: HplmnAlgoSwitch=INTRA_RAT_HO_WITH_HPLMN_SW-1;
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=BACK_TO_HPLMN_INTER_FREQ_HO_SW-1;
//Configuring the threshold for event A4 related to RFSP-based inter-frequency handover back to the
HPLMN
MOD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=2, BackToHplmnHoA5RsrpThld2=-104;
//Configuring the threshold for event A5 related to RFSP-based inter-frequency handover back to the
HPLMN
MOD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=2, BackToHplmnHoA5RsrpThld1=-43,
BackToHplmnHoA5RsrpThld2=-104;
//Configuring the retry period of the RFSP-based inter-frequency handover back to the HPLMN
MOD GNBFRFSPCONFIG: OperatorId=0, RfspIndex=2, BackToHplmnHoRetryPeriod=90;
```

- **RFSP-based unnecessary handover policy**
 //The following describes the activation command example with one RFSP index added for an operator.
 //Adding gNodeB RFSP configurations
 ADD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3;
 //Turning on the Unnecessary Handover Suppression Switch for RFSP index 3 because unnecessary handovers are not required for UEs with RFSP index 3
 MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3,
 RfspAlgoSwitch=UNNECESSARY_HO_SUPPRESS_SW-1;

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- **RFSP-based cell reselection policy and basic RFSP-based handover policy**

TDD configuration examples

```
//Running the following cell-level command to deactivate the RFSP-specific frequency-priority-based
inter-frequency handover function for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=FREQ_PRIORITY_BASED_HO_SW-0;
//Running the following cell-level command to deactivate the RFSP-based NR inter-frequency
handover policy for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=INTER_FREQ_LIST_CTRL_SW-0;
//Running the following cell-level command to deactivate the RFSP-based NG-RAN-to-E-UTRAN
handover policy for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=NR_TO_EUTRAN_FREQ_LIST_CTRL_SW-0;
//Running the following base-station-level command to remove the binding relationship between
RFSP indexes and frequency priority groups to deactivate all RFSP-based cell reselection and handover
policies for all cells of the base station
MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=2, gNBFreqPriorityGroupId=255;
MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3, gNBFreqPriorityGroupId=255;
```

- **Policy of RFSP-based inter-frequency handover back to the HPLMN**

TDD configuration examples

```
//Running the following cell-level command to deactivate the policy of RFSP-based inter-frequency
handover back to the HPLMN for the current cell (NrCellId=0)
MOD NRCELLALGOSWITCH: NrCellId=0, RfspAlgoSwitch=BACK_TO_HPLMN_INTER_FREQ_HO_SW-0;
//Running the following base-station-level command to remove the binding relationship between
RFSP indexes of roaming UEs and frequency priority groups to deactivate the policy of RFSP-based
inter-frequency handover back to the HPLMN for all cells of the base station
MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=2, gNBFreqPriorityGroupId=255;
```

- **RFSP-based unnecessary handover policy**
 //Turning off the Unnecessary Handover Suppression Switch for RFSP index 3 because unnecessary handovers are required for UEs with RFSP index 3
 MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3,
 RfspAlgoSwitch=UNNECESSARY_HO_SUPPRESS_SW-0;

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

RFSP-based Cell Reselection Policy

The following uses a reselection to an NR cell in SA networking as an example to describe the verification procedure.

- Step 1** Enable a UE to access a cell.
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.
- Step 3** Do not use the UE to perform any service or enable it to release all services. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The RFSP-based cell reselection policy has taken effect if the NR frequencies contained in the *cellReselectionPriorities* IE of the RRCRelease message delivered by the gNodeB are the non-serving frequencies which (1) are in the gNodeB frequency priority group configured using the **ADD GNBFREQPRIORITYGROUP** command and (2) serve the neighboring cells of the local cell.

----End

The following uses a reselection to an E-UTRAN cell in SA networking as an example to describe the verification procedure.

- Step 1** Enable a UE to access a cell.
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.
- Step 3** Do not use the UE to perform any service or enable it to release all services. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The RFSP-based cell reselection policy has taken effect if the E-UTRAN frequencies contained in the *cellReselectionPriorities* IE of the RRCRelease message delivered by the gNodeB are the E-UTRAN frequencies which (1) are in the gNodeB frequency priority group configured using the **ADD GNBFREQPRIORITYGROUP** command and (2) serve the neighboring cells of the local cell.

----End

Basic RFSP-based Handover Policy

The following uses a handover to an NR cell in SA networking as an example to describe the verification procedure.

Step 1 Enable a UE to access a cell.

Step 2 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.

Step 3 Use the UE to perform services, and then trigger an intra-NR handover for the UE. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The RFSP-based handover policy has taken effect if the to-be-measured frequencies contained in inter-frequency event A4 or A5 measurement configurations in the RRCReconfiguration message delivered by the gNodeB are the non-serving frequencies which (1) are in the gNodeB frequency priority group configured using the **ADD GNBREQPRIORITYGROUP** command and (2) serve the neighboring cells of the local cell.

----End

The following uses a handover to an E-UTRAN cell in SA networking as an example to describe the verification procedure.

Step 1 Enable a UE to access a cell.

Step 2 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.

Step 3 Use the UE to perform services, and then trigger an NG-RAN-to-E-UTRAN handover for the UE. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The RFSP-based handover policy has taken effect if the to-be-measured frequencies contained in inter-frequency event B1 or B2 measurement configurations in the RRCReconfiguration message delivered by the gNodeB are the E-UTRAN frequencies which (1) are in the gNodeB frequency priority group configured using the **ADD GNBREQPRIORITYGROUP** command and (2) serve the neighboring cells of the local cell.

----End

The following uses an inter-frequency SCell change in NSA networking as an example to describe the verification procedure.

Step 1 Enable a UE to access a cell.

Step 2 On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **X2 Interface Trace**. In the displayed **X2 Interface Trace** dialog box, check whether the SgNB Addition Request message contains the *subscriberProfileIDforRFP* IE. If it does, the core network has delivered the SPID information, which is equivalent to the RFSP information in the NG-RAN.

- Step 3** Use the UE to perform services and trigger an inter-frequency SCell change for the UE. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The RFSP-based SCell change policy has taken effect if the to-be-measured frequencies contained in inter-frequency event A4 or A5 measurement configurations in the RRC Connection Reconfiguration message delivered by the eNodeB are the non-serving frequencies which (1) are in the gNodeB frequency priority group configured using the **ADD GNBREQPRIORITYGROUP** command and (2) serve the neighboring cells of the local secondary cell.

----End

Policy of RFSP-based Inter-Frequency Handover Back to the HPLMN

You can observe whether the policy of RFSP-based inter-frequency handover back to the HPLMN is activated by observing counters or tracing signaling messages.

Observe relevant counters. If any of the following counters returns a value other than 0, the policy of RFSP-based inter-frequency handover back to the HPLMN has taken effect.

- **N.HO.InterFreq.BackToHPLMN.PrepAttOut**
- **N.HO.InterFreq.BackToHPLMN.ExecAttOut**
- **N.HO.InterFreq.BackToHPLMN.ExecSuccOut**

Signaling tracing: The following uses a handover of a roaming UE back to an NR cell on the frequency of its HPLMN in SA networking as an example to describe the verification procedure.

- Step 1** Enable the roaming UE to access a cell (The cell is not an NR cell on the frequency of the UE's HPLMN).
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.
- Step 3** Use the UE to perform services, and then trigger an intra-NR handover for the UE. On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the displayed **Uu Interface Trace** dialog box. The policy of RFSP-based inter-frequency handover back to the HPLMN has taken effect if the to-be-measured frequencies contained in inter-frequency event A4 or A5 measurement configurations in the RRCReconfiguration message delivered by the gNodeB are the non-serving frequencies which (1) have **gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag** set to **TRUE** by using **ADD GNBREQPRIORITYGROUP** and (2) serve the neighboring cells of the local cell.

----End

RFSP-based Unnecessary Handover Policy

This basic function ensures basic service performance. Separate activation verification is not required.

4.4.3 Network Monitoring

Currently, there is no RFSP-level monitoring method for RFSP-based camping and handover policies. The network-level monitoring methods are as follows:

- The network-level monitoring methods for policies of RFSP-based cell reselections (within the NR system) and handovers (within the NR system) are the same as those for cell reselections and handovers described in *SA Mobility Management in Idle Mode* and *SA Mobility Management in Connected Mode*.
- The network-level monitoring methods for policies of RFSP-based cell reselections (from the NR system to E-UTRAN) and handovers (from the NR system to E-UTRAN) are the same as those for cell reselections and handovers described in *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*, *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*, and *NSA-SA Selection Based on User Experience*.

RFSP-based unnecessary handover policy: When there are RFSP-specific UEs on the network, you can monitor the gains and impacts of this function by observing the counters listed in [4.2 Network Analysis](#).

5 RFSP-based DRX Policy

5.1 Principles

When DRX is enabled, the RFSP-based DRX policy can be used. In this way, UEs with different power saving or delay requirements can use RFSP-dedicated DRX parameters to perform DRX.

 NOTE

This document describes only the content exclusive to RFSP-based DRX compared with basic DRX functions. For details about basic DRX functions, see *DRX*.

The following describes how the RFSP-specific DRX parameters are configured and delivered to UEs on the gNodeB side as well as how these parameters are used by UEs for DRX:

1. Configure DRX parameter groups on the gNodeB side.
- On the CU side

The **gNBDrxParamGroup** MO is used to configure DRX parameter groups on the CU side. [Table 5-1](#) describes the major parameters.

Table 5-1 DRX parameter group on the CU side

Parameter Name	Parameter ID	Description
DRX Parameter Group ID	gNBDrxParamGroup.DrxParamGroupId	This parameter specifies the ID of a DRX parameter group on the CU side.
Long Cycle	gNBDrxParamGroup.LongCycle	This parameter specifies the length of a long DRX cycle. A long DRX cycle saves more power for UEs.

Parameter Name	Parameter ID	Description
Short Cycle	gNBDrxParamGroup.ShortCycle	This parameter specifies the length of a short DRX cycle. A short DRX cycle enables UEs to obtain better service experience.
Short Cycle Timer	gNBDrxParamGroup.ShortCycleTimer	This parameter specifies the number of consecutive repetitions of a short DRX cycle.

- On the DU side

The **gNBDuDrxParamGroup** MO is used to configure DRX parameter groups on the DU side. [Table 5-2](#) describes the major parameters.

Table 5-2 DRX parameter group on the DU side

Parameter Name	Parameter ID	Description
DU DRX Parameter Group ID	gNBDUDrxParamGroup.DuDrxParamGroupId	This parameter specifies the ID of a DRX parameter group on the DU side.
On Duration Timer	gNBDUDrxParamGroup.OnDurationTimer	This parameter specifies the length of the On Duration Timer.
DRX Inactivity Timer	gNBDUDrxParamGroup.DrxInactivityTimer	This parameter specifies the length of the DRX Inactivity Timer.
DRX Retransmission Timer	gNBDUDrxParamGroup.DrxRetransTimer	This parameter specifies the length of the DRX downlink and uplink retransmission timers.

This function involves the **gNBDrxParamGroup.DrxParamGroupId** and **gNBDUDrxParamGroup.DuDrxParamGroupId** parameters that need to be bound with RFSP indexes. For details about other DRX parameters, see *DRX*.

2. The gNodeB obtains the RFSP indexes of UEs from the core network.
 3. Based on the obtained RFSP indexes, bind the RFSP indexes to the configured DRX parameter groups on the gNodeB to generate the RFSP-specific DRX parameters.
- On the CU side

The **gNBRfspConfig** MO is used for the binding. [Table 5-3](#) lists the related parameters.

Table 5-3 Parameters for binding an RFSP index to a CU-side DRX parameter group

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of the UE.
DRX Parameter Group ID	gNBRfspConfig.DrxParamGroupId	This parameter specifies the ID of the DRX parameter group on the CU side bound to the RFSP index. This parameter has a mapping relationship with the gNBDrxParamGroup.DrxParamGroupId parameter. If this parameter is set to 255 , the CU-side DRX parameter group bound to the RFSP index is invalid. Otherwise, the parameter group is valid.

- On the DU side

The **gNBDURfspConfig** MO is used for the binding. [Table 5-4](#) describes the related parameters.

Table 5-4 Parameters for binding an RFSP index to a DU-side DRX parameter group

Parameter Name	Parameter ID	Description
Operator ID	gNBDURfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBDURfspConfig.RfspIndex	This parameter specifies the RFSP index of the UE.

Parameter Name	Parameter ID	Description
DU DRX Parameter Group ID	gNBDURfspConfig.DuDrxParamGroupId	This parameter specifies the ID of the DRX parameter group on the DU side bound to the RFSP index. This parameter has a mapping relationship with the gNBDUDrxParamGroup.DuDrxParamGroupId parameter. If this parameter is set to 255, the DU-side DRX parameter group bound to the RFSP index is invalid. Otherwise, the parameter group is valid.

4. The gNodeB checks UEs' RFSP indexes and determines whether DRX takes effect.
- Scenario 1: If the parameter settings listed in [Table 5-5](#) are used, the RFSP-based DRX policy takes effect for UEs with the specified RFSP index. In this case, the process proceeds to [4](#).

Table 5-5 Scenario 1

Parameter Name	Parameter ID	Description
DRX Not Configured Switch	NOT_CONFIG_DRX_SW option of the gNBRfspConfig.RfspAlgoSwitch parameter	This switch is turned off for the corresponding RFSP index.

Parameter Name	Parameter ID	Description
DRX Parameter Group ID	gNBRfspConfig.DrxParamGroupId	The DRX parameter groups bound to the corresponding RFSP index are used on both the CU and DU sides when all of the following conditions are met: • The corresponding RFSP index is bound to a valid DRX parameter group on the CU side. • The corresponding RFSP index is bound to a valid DRX parameter group on the DU side. When all of the following conditions are met, the CU-side DRX parameter group bound to the corresponding RFSP index is used on the CU side, and the DU-side cell-level DRX parameter group is used on the DU side: • The corresponding RFSP index is bound to a valid DRX parameter group on the CU side. • The corresponding RFSP index is bound to an invalid DRX parameter group on the DU side. When all of the following conditions are met, the CU-side cell-level DRX parameter group is used on the CU side, and the DU-side DRX parameter group bound to the corresponding RFSP index is used on the DU side: • The corresponding RFSP index is bound to an invalid DRX parameter group on the CU side. • The corresponding RFSP index is bound to a valid DRX parameter group on the DU side.
DU DRX Parameter Group ID	gNBDURfspConfig.DuDrxParamGroupId	

 NOTE

It is recommended that valid DRX parameter groups be bound on both the CU and DU sides for an RFSP index.

If the status of the DRX Not Configured Switch changes, this change takes effect immediately for newly accessed UEs but not for online UEs.

- Scenario 2: If the parameter settings listed in [Table 5-6](#) are used, the cell-level DRX policy takes effect for UEs with the specified RFSP index. That is, cell-level DRX parameter groups are used on both the CU and DU sides. The process ends. The cell-level DRX policy indicates the QCI-based DRX parameter configuration. For details, see *DRX*.

Table 5-6 Scenario 2

Parameter Name	Parameter ID	Description
DRX Not Configured Switch	NOT_CONFIG_DRX_SW option of the gNBRfspConfig.RfspAlgoSwitch parameter	This switch is turned off for the corresponding RFSP index.
DRX Parameter Group ID	gNBRfspConfig.DrxParamGroupId	The cell-level DRX parameter group is used on both the CU and DU sides when all of the following conditions are met: • The corresponding RFSP index is bound to an invalid DRX parameter group on the CU side. • The corresponding RFSP index is bound to an invalid DRX parameter group on the DU side.
DU DRX Parameter Group ID	gNBDURfspConfig.DuDrxParamGroupId	

- Scenario 3: If the parameter settings listed in [Table 5-7](#) are used, neither the RFSP-based DRX policy nor the cell-level DRX policy takes effect for UEs with the specified RFSP index. The process ends.

Table 5-7 Scenario 3

Parameter Name	Parameter ID	Description
DRX Not Configured Switch	NOT_CONFIG_DRX_SW option of the gNBRRfspConfig.RfspAlgoSwitch parameter	This switch is turned on for the corresponding RFSP index.

5. The gNodeB delivers the RFSP-specific DRX parameters to a UE using the RRCReconfiguration message.
6. The UE obtains the RFSP-specific DRX parameters and checks the RFSP index configuration in the RFSP-specific DRX parameters to determine the way how the DRX policy takes effect.
 - If the RFSP index of the UE has a corresponding **gNBRRfspConfig.RfspIndex**, the configuration of RFSP-specific DRX parameters for this RFSP index is checked.
 - If the RFSP index is bound to a valid DRX parameter group on the CU or DU side, RFSP-based DRX can take effect on the UE.
The DRX parameter group bound to the corresponding RFSP index is used on both the CU and DU sides when all of the following conditions are met:
 - The corresponding RFSP index is bound to a valid DRX parameter group on the CU side.
 - The corresponding RFSP index is bound to a valid DRX parameter group on the DU side.
 When all of the following conditions are met, the CU-side DRX parameter group bound to the corresponding RFSP index is used on the CU side, and the DU-side cell-level DRX parameter group is used on the DU side:
 - The corresponding RFSP index is bound to a valid DRX parameter group on the CU side.
 - The corresponding RFSP index is bound to an invalid DRX parameter group on the DU side.
 When all of the following conditions are met, the CU-side cell-level DRX parameter group is used on the CU side, and the DU-side DRX parameter group bound to the corresponding RFSP index is used on the DU side:
 - The corresponding RFSP index is bound to an invalid DRX parameter group on the CU side.
 - The corresponding RFSP index is bound to a valid DRX parameter group on the DU side.
 - Otherwise, RFSP-based DRX policy does not take effect for the UE.
 - If the RFSP index of the UE does not have a corresponding **gNBRRfspConfig.RfspIndex**, the configuration of RFSP-specific DRX parameters for RFSP index 0 is checked.

- If there is a **gNBRfspConfig.RfspIndex** parameter with the value being **0** and this index is bound to a valid DRX parameter group on the CU or DU side, RFSP-based DRX can take effect on the UE.

The DRX parameter groups bound to RFSP index 0 are used on both the CU and DU sides when all of the following conditions are met:

- RFSP index 0 is bound to a valid DRX parameter group on the CU side.
- RFSP index 0 is bound to a valid DRX parameter group on the DU side.

When all of the following conditions are met, the CU-side DRX parameter group bound to the RFSP index 0 is used on the CU side, and the DU-side cell-level DRX parameter group is used on the DU side:

- RFSP index 0 is bound to a valid DRX parameter group on the CU side.
- RFSP index 0 is bound to an invalid DRX parameter group on the DU side.

When all of the following conditions are met, the CU-side cell-level DRX parameter group is used on the CU side, and the DU-side DRX parameter group bound to RFSP index 0 is used on the DU side:

- RFSP index 0 is bound to an invalid DRX parameter group on the CU side.
- RFSP index 0 is bound to a valid DRX parameter group on the DU side.

- Otherwise, RFSP-based DRX policy does not take effect for the UE.

5.2 Network Analysis

5.2.1 Benefits

This function enables operators to customize DRX policies for different types of UEs to improve user experience.

If not configuring DRX for UEs with specific RFSP indexes takes effect, UEs with specific RFSP indexes will not enter DRX mode. In this case, the average uplink and downlink throughput of these UEs increases and their service delay decreases.

5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

If not configuring DRX for UEs with specific RFSP indexes takes effect, UEs with specific RFSP indexes will not enter DRX mode, increasing their power consumption.

5.3 Requirements

5.3.1 Licenses

There are no license requirements for basic functions.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	DRX	BASIC_DRX_SW option of the NRDUCellUePwrSaving.NrDuCellDrxAlgoSwitch parameter	<i>DRX</i>	The RFSP-based DRX policy function requires the DRX function to be enabled.

5.3.2.2 Mutually Exclusive Functions

None

5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-RAT ANR	NR_NR_ANR_SW option of the NRCellAlgoSwitch . <i>AnrSwitch</i> parameter	<i>ANR</i>	• UEs can perform reportCGI measurements only after entering the DRX sleep time. • When the gNodeB triggers UEs to read the NCGIs, the UEs automatically use a long DRX cycle. The setting of the NOT_CONFIG_DRX_SW option of the gNBRfspConfig.<i>RfspAlgoSwitch</i> parameter does not affect the configuration of the long DRX cycle initiated by intra-RAT ANR.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-RAT ANR	NR_EUTRAN_ANR_SW option of the NRCellAlgoSwitch . AnrSwitch parameter	<i>ANR</i>	• UEs can perform reportCGI measurements only after entering the DRX sleep time. • When the gNodeB triggers UEs to read the ECGIs, the UEs automatically use a long DRX cycle. The setting of the NOT_CONFIG_DRX_SW option of the gNBRfspConfig.RfspAlgoSwitch parameter does not affect the configuration of the long DRX cycle initiated by inter-RAT ANR.
High-frequency TDD	None	None	None	None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite support the function only in low-frequency scenarios.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

5.3.4 Others

- UEs
UEs must comply with 3GPP Release 15 or later.
- Core network
The core network must comply with 3GPP Release 15 or later, and the AMF must support RFSP configuration.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

[Table 5-8](#) describes the parameters used for function activation.

Table 5-8 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
DRX Parameter Group ID	gNBDrxParamGroup.DrxParamGroupId	Set this parameter based on the network plan.
Long Cycle	gNBDrxParamGroup.LongCycle	Set this parameter based on the network plan.
DU DRX Parameter Group ID	gNBDUDrxParamGroup.DuDrxParamGroupId	Set this parameter based on the network plan.
On Duration Timer	gNBDUDrxParamGroup.OnDurationTimer	Set this parameter based on the network plan.
DRX Inactivity Timer	gNBDUDrxParamGroup.DrxInactivityTimer	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
DRX Retransmission Timer	gNBDUDrxParamGroup.DrxRetransTimer	Set this parameter based on the network plan.
Operator ID	gNBRfspConfig.OperatorId	Set this parameter based on the network plan.
Index to RAT/Frequency Selection Priority	gNBRfspConfig.RfspIndex	Set this parameter based on the network plan.
DRX Parameter Group ID	gNBRfspConfig.DrxParamGroupId	Set this parameter based on the network plan.
RFSP Algorithm Switch	gNBRfspConfig.RfspAlgoSwitch	If the throughput and service delay of UEs with a specific RFSP index need to be preferentially guaranteed, select the NOT_CONFIG_DRX_SW option of this parameter to disable DRX for these UEs. If power saving needs to be preferentially guaranteed, deselect the NOT_CONFIG_DRX_SW option of this parameter to enable DRX for UEs with the specified RFSP index.
Operator ID	gNBDURfspConfig.OperatorId	Set this parameter based on the network plan.
Index to RAT/Frequency Selection Priority	gNBDURfspConfig.RfspIndex	Set this parameter based on the network plan.
DU DRX Parameter Group ID	gNBDURfspConfig.DuDrxParamGroupId	Set this parameter based on the network plan.

 NOTE

It is recommended that valid DRX parameter groups be bound on both the CU and DU sides for an RFSP index.

If the status of the DRX Not Configured Switch changes, this change takes effect immediately for newly accessed UEs but not for online UEs.

5.4.1.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring DRX parameter groups (including CU DRX parameter groups and DU DRX parameter groups)
ADD GNBDRXPARAMGROUP: DrxParamGroupId=0, LongCycle=MS320;
ADD GNBDRXPARAMGROUP: DrxParamGroupId=1, LongCycle=MS160;
ADD GNBUDRXPARAMGROUP: DuDrxParamGroupId=0, OnDurationTimer=MS10,
DrxInactivityTimer=MS100, DrxRetransTimer=SL16;
ADD GNBUDRXPARAMGROUP: DuDrxParamGroupId=1, OnDurationTimer=MS10,
DrxInactivityTimer=MS80, DrxRetransTimer=SL16;
//Binding RFSP indexes to the configured CU DRX parameter groups
ADD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=2, DrxParamGroupId=0;
ADD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=3, DrxParamGroupId=1;
//Binding RFSP indexes to the configured DU DRX parameter groups
ADD GNBUDRFSPCONFIG: OperatorId=0, RfspIndex=2, DuDrxParamGroupId=0;
ADD GNBUDRFSPCONFIG: OperatorId=0, RfspIndex=3, DuDrxParamGroupId=1;
//Enabling the function of not configuring DRX for UEs with RFSP index 2 to preferentially guarantee the
throughput and service delay of these UEs
MOD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=2, RfspAlgoSwitch=NOT_CONFIG_DRX_SW-1;
//Disabling the function of not configuring DRX for UEs with RFSP index 3 to preferentially guarantee the
power saving performance of these UEs
MOD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=3, RfspAlgoSwitch=NOT_CONFIG_DRX_SW-0;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Deactivating the RFSP-based DRX policy for all cells of a base station
//1. Removing the binding relationship between the RFSP indexes and the configured CU DRX parameter
groups
MOD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=2, DrxParamGroupId=255;
MOD GNBDRFSPCONFIG: OperatorId=0, RfspIndex=3, DrxParamGroupId=255;
//2. Removing the binding relationship between the RFSP indexes and the configured DU DRX parameter
groups
MOD GNBUDRFSPCONFIG: OperatorId=0, RfspIndex=2, DuDrxParamGroupId=255;
MOD GNBUDRFSPCONFIG: OperatorId=0, RfspIndex=3, DuDrxParamGroupId=255;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

The following uses the RFSP-based DRX policy customized for UEs in SA networking as an example to describe the verification method:

- Step 1** Enable a UE to access the network and ensure that the conditions for entering DRX mode are met.
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.
- Step 3** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the **Uu Interface Trace** dialog box. If the RRCReconfiguration message contains DRX parameters that are consistent with the configured RFSP-based DRX parameters, the UE has entered the DRX mode based on the RFSP-based DRX policy.

----End

The following uses the SPID-specific DRX policy customized for UEs in NSA networking as an example to describe the verification method:

- Step 1** Enable a UE to access the network and ensure that the conditions for entering DRX mode are met.
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **X2 Interface Trace**. In the displayed **X2 Interface Trace** dialog box, check whether the SgNB Addition Request message contains the *subscriberProfileIDforRFP* IE. If it does, the core network has delivered the SPID information, which is equivalent to the RFSP information in the NG-RAN.
- Step 3** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. Observe the **Uu Interface Trace** dialog box. If the RRC Connection Reconfiguration message delivered by the eNodeB contains DRX parameters that are consistent with the configured SPID-specific DRX parameters, the UE has entered the DRX mode based on the SPID-specific DRX policy.

----End

NOTE

For details about the conditions for entering DRX mode, see *DRX*.

5.4.3 Network Monitoring

Currently, there is no RFSP-level monitoring method for the RFSP-based DRX policy. The network-level monitoring methods are the same as those for DRX. For details, see *DRX*.

6 RFSP-based VoNR Policy

6.1 Principles

If basic VoNR function switches are turned on, the gNodeB can determine whether VoNR takes effect for specific UEs based on RFSP indexes through the RFSP-based VoNR policy.

 NOTE

This document describes the content exclusive to the RFSP-based VoNR policy compared with basic VoNR functions. For details about basic VoNR functions, see *VoNR*.

The general principle of the RFSP-based VoNR policy is as follows: The gNodeB differentially configures the **VoNR Prohibition Switch (VONR_PROHIBIT_SW)** option of the **gNBRfspConfig.RfspAlgoSwitch** parameter) for UEs with different RFSP indexes so that differentiated VoNR policies can take effect for these UEs.

Specifically:

1. The gNodeB obtains the RFSP indexes of UEs from the core network.
2. Based on the obtained RFSP indexes, the gNodeB configures the **VoNR Prohibition Switch** differentially for RFSP indexes. Then, the gNodeB determines whether to enable VoNR to take effect for the UEs with specified RFSP indexes based on the setting of this option.
 - a. The **VoNR Prohibition Switch** is configured differentially for RFSP indexes.

Parameter Name	Parameter ID	Description
Operator ID	gNBRfspConfig.OperatorId	This parameter specifies the operator of a UE.
Index to RAT/ Frequency Selection Priority	gNBRfspConfig.RfspIndex	This parameter specifies the RFSP index of a UE.

Parameter Name	Parameter ID	Description
VoNR Prohibition Switch	VONR_PROHIBIT_SW option of the gNB RfspConfig.RfspAlgoSwitch parameter	This option specifies whether to enable VoNR to take effect for the UEs with specified RFSP indexes.

- b. The gNodeB determines whether to enable VoNR to take effect for the UEs with specified RFSP indexes.
 - If the **VoNR Prohibition Switch** is turned off for an RFSP index, VoNR can take effect for UEs with the RFSP index.
 - If the **VoNR Prohibition Switch** is turned on for an RFSP index, VoNR cannot take effect for UEs with the RFSP index.

6.2 Network Analysis

6.2.1 Benefits

With the RFSP-based VoNR policy, operators can customize VoNR policies for different types of UEs to improve service experience of these UEs.

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).

After the RFSP-based VoNR policy takes effect, VoNR does not take effect for UEs with the specified RFSP index. As a result, the connection delay of voice services initiated by these UEs increases, and the values of counters related to 5QI 1 services decrease. For details about the counters related to 5QI 1 services, see *VoNR*.

6.3 Requirements

6.3.1 Licenses

There are no license requirements for basic functions.

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

6.3.2.1 Prerequisite Functions

None

6.3.2.2 Mutually Exclusive Functions

None

6.3.2.3 Function Impacts

None

6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite support the function only in low-frequency scenarios.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

6.3.4 Others

- UEs
UEs must comply with 3GPP Release 15 or later.
- Core network
The core network must comply with 3GPP Release 15 or later, and the AMF must support RFSP configuration.

6.4 Operation and Maintenance

6.4.1 Data Configuration

6.4.1.1 Data Preparation

[Table 6-1](#) describes the parameters used for function activation.

Table 6-1 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Operator ID	gNB RFSP Config. <i>OperatorId</i>	Set this parameter based on the network plan.
Index to RAT/ Frequency Selection Priority	gNB RFSP Config. <i>RfspIndex</i>	Set this parameter based on the network plan.
RFSP Algorithm Switch	gNB RFSP Config. <i>RfspAlgoSwitch</i>	If VoNR is required for the UEs with a specified RFSP index, it is recommended that the VONR_PROHIBIT_SW option be deselected for the RFSP index. If VoNR is not required for these UEs, it is recommended that the VONR_PROHIBIT_SW option be selected for the RFSP index.

6.4.1.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//The following describes the activation command example with one RFSP index added for an operator.  
//Adding gNodeB RFSP configurations  
ADD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3;  
//Turning on the VoNR Prohibition Switch for RFSP index 3 if VoNR is not required for UEs with this RFSP  
index  
MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3, RfspAlgoSwitch=VONR_PROHIBIT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Turning off the VoNR Prohibition Switch for RFSP index 3 if VoNR is required for UEs with this RFSP index  
MOD GNB RFSP CONFIG: OperatorId=0, RfspIndex=3, RfspAlgoSwitch=VONR_PROHIBIT_SW-0;
```

6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

6.4.2 Activation Verification

The RFSP-based VoNR policy in SA networking can be verified as follows:

- Step 1** Enable a UE to access the network and ensure that the conditions for VoNR to take effect are met.
- Step 2** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **NG Interface Trace**. In the displayed **NG Interface Trace** dialog box, check whether the INITIAL CONTEXT SETUP REQ message contains the *IndexToRFSP* IE. If it does, the core network has delivered the RFSP information.
- Step 3** On the MAE-Access, choose **Monitor > Signaling Trace > Signaling Trace Management**. In the navigation tree on the left, choose **Uu Interface Trace**. In the displayed **Uu Interface Trace** dialog box, check whether 5QI 1 bearers are set up. If they are not, the RFSP-based VoNR policy has taken effect for the UEs.

-----End

6.4.3 Network Monitoring

At present, there is no RFSP-level monitoring method for the RFSP-based VoNR policy. The network-level monitoring methods are the same as those for VoNR. For details, see *VoNR*.

7 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

8 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

9 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

10 Reference Documents

- 3GPP TS 38.304: "NR; User Equipment (UE) procedures in Idle mode and RRC"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *SA Mobility Management in Idle Mode*
- *SA Mobility Management in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *NSA-SA Selection Based on User Experience*
- *DRX*
- *Mobility Load Balancing in Idle Mode*
- *VoNR*
- *ANR*
- *RedCap*